

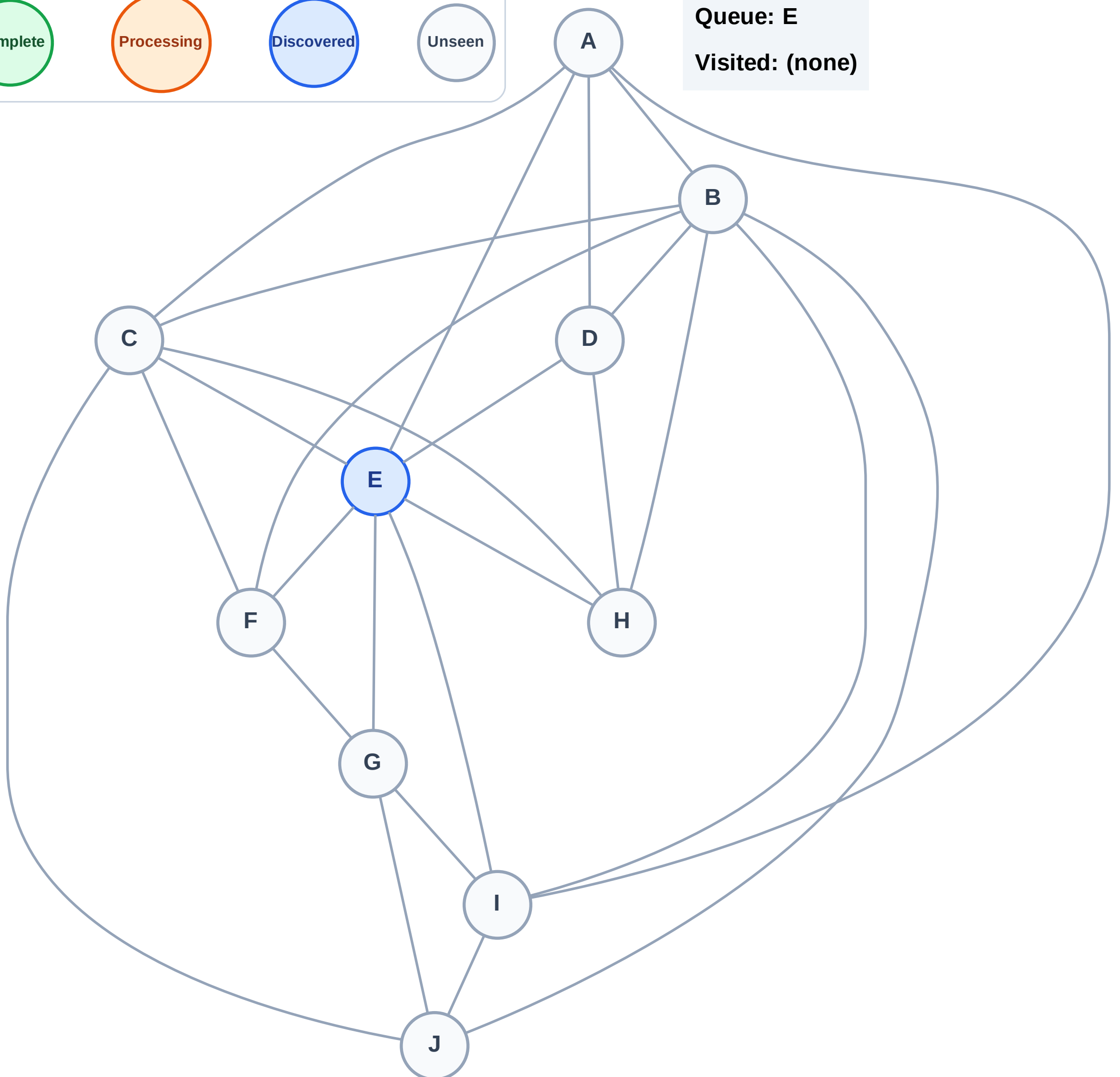
BFS Traversal

Step 1: Start at E

Legend



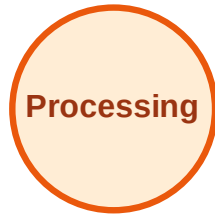
Queue: E
Visited: (none)



BFS Traversal

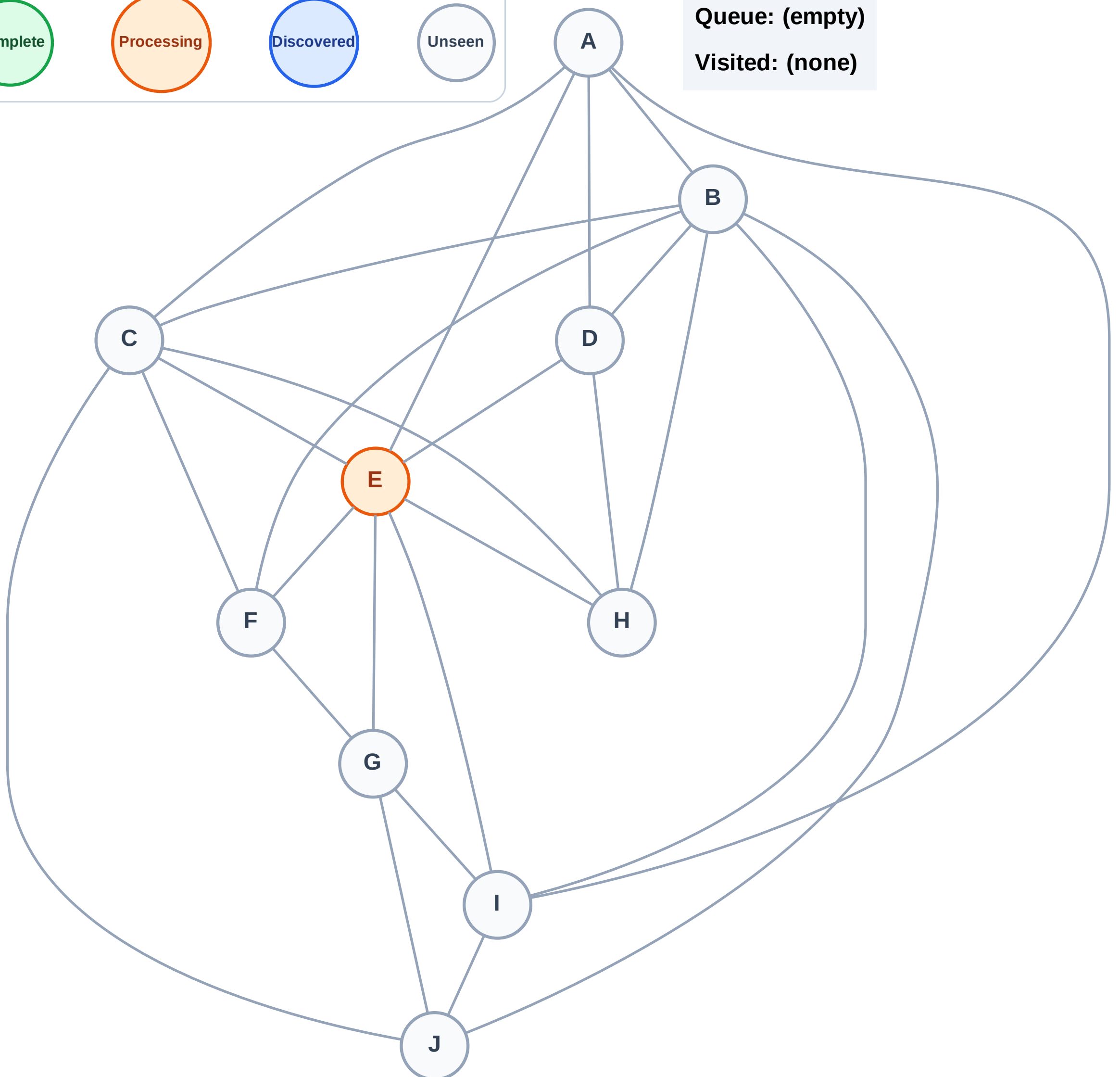
Step 2: Process node E

Legend



Queue: (empty)

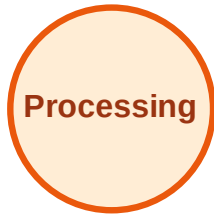
Visited: (none)



BFS Traversal

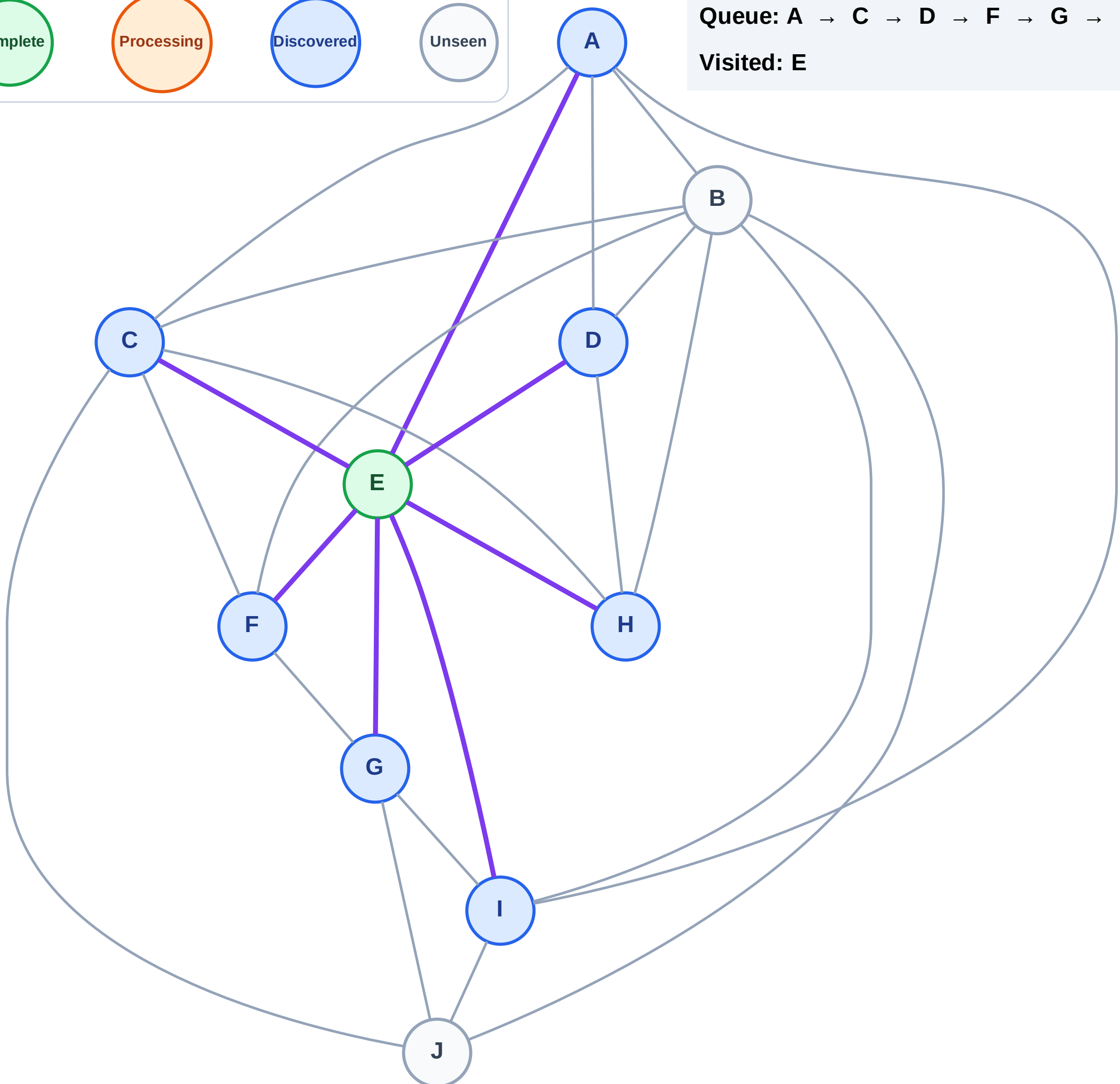
Step 3: Discover A, C, D, F, G, H, I from E

Legend



Queue: A → C → D → F → G → H → I

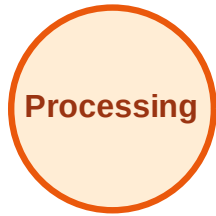
Visited: E



BFS Traversal

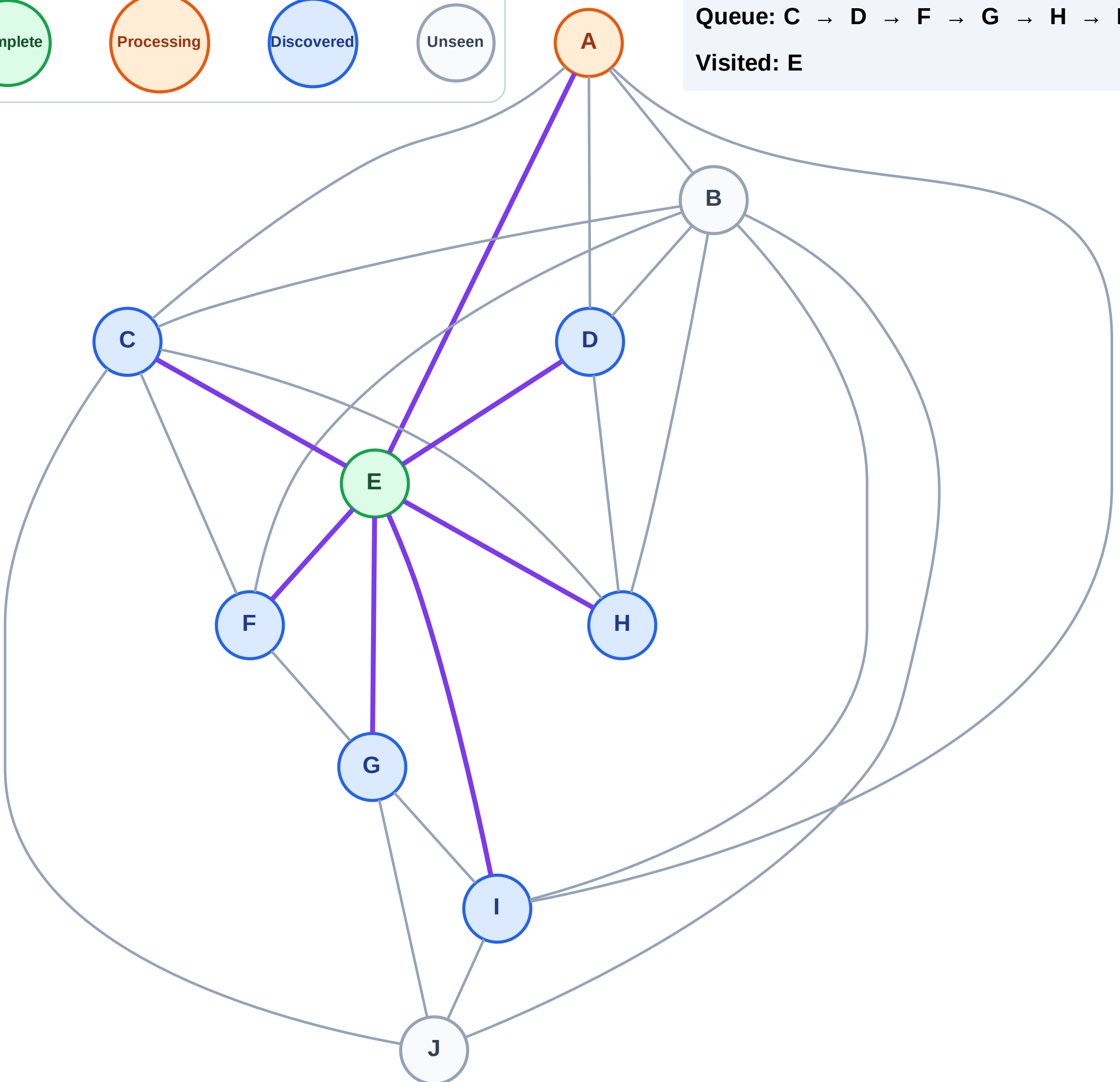
Step 4: Process node A

Legend



Queue: C → D → F → G → H → I

Visited: E



BFS Traversal

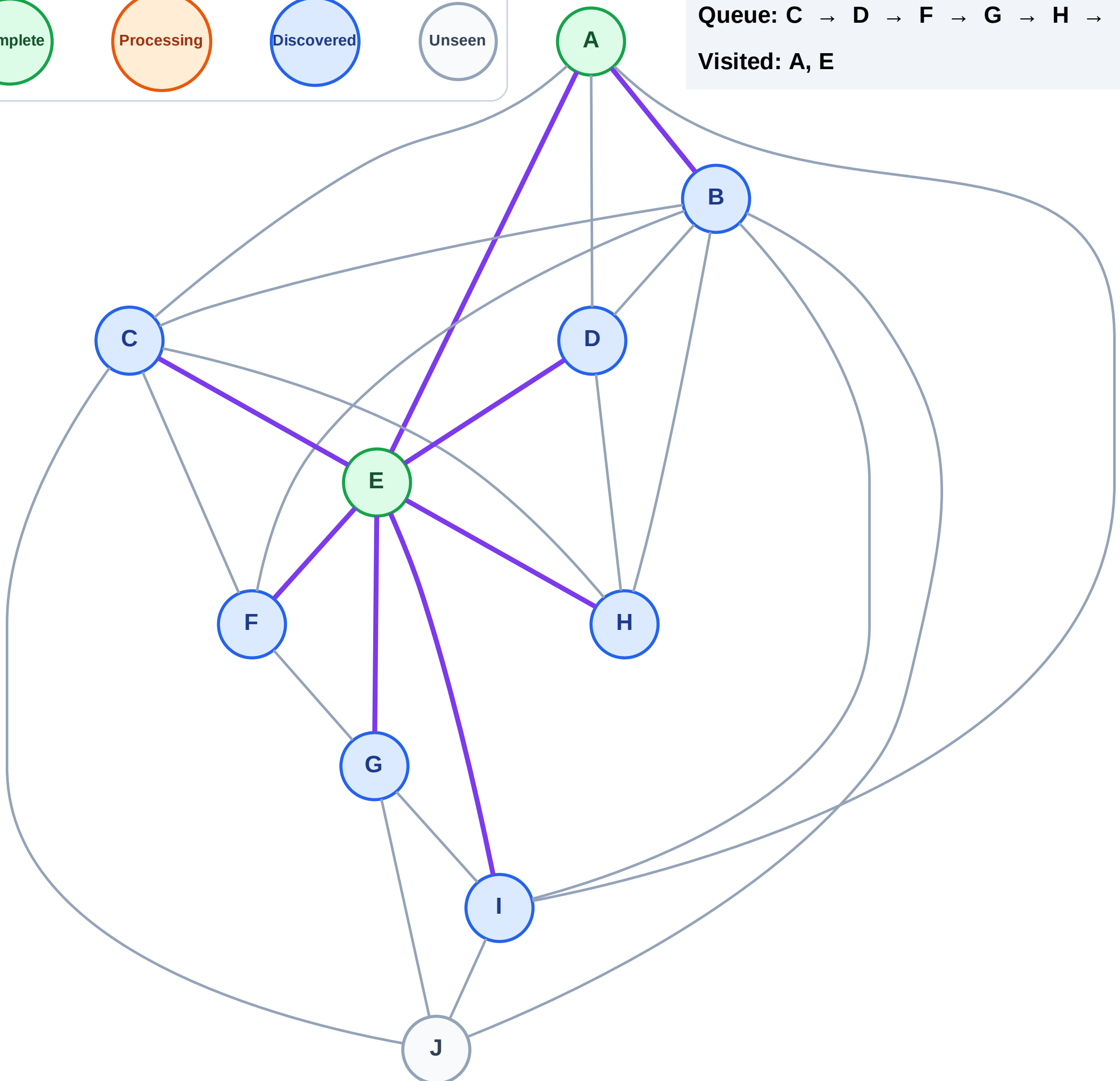
Step 5: Discover B from A

Legend



Queue: C → D → F → G → H → I → B

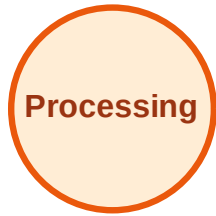
Visited: A, E



BFS Traversal

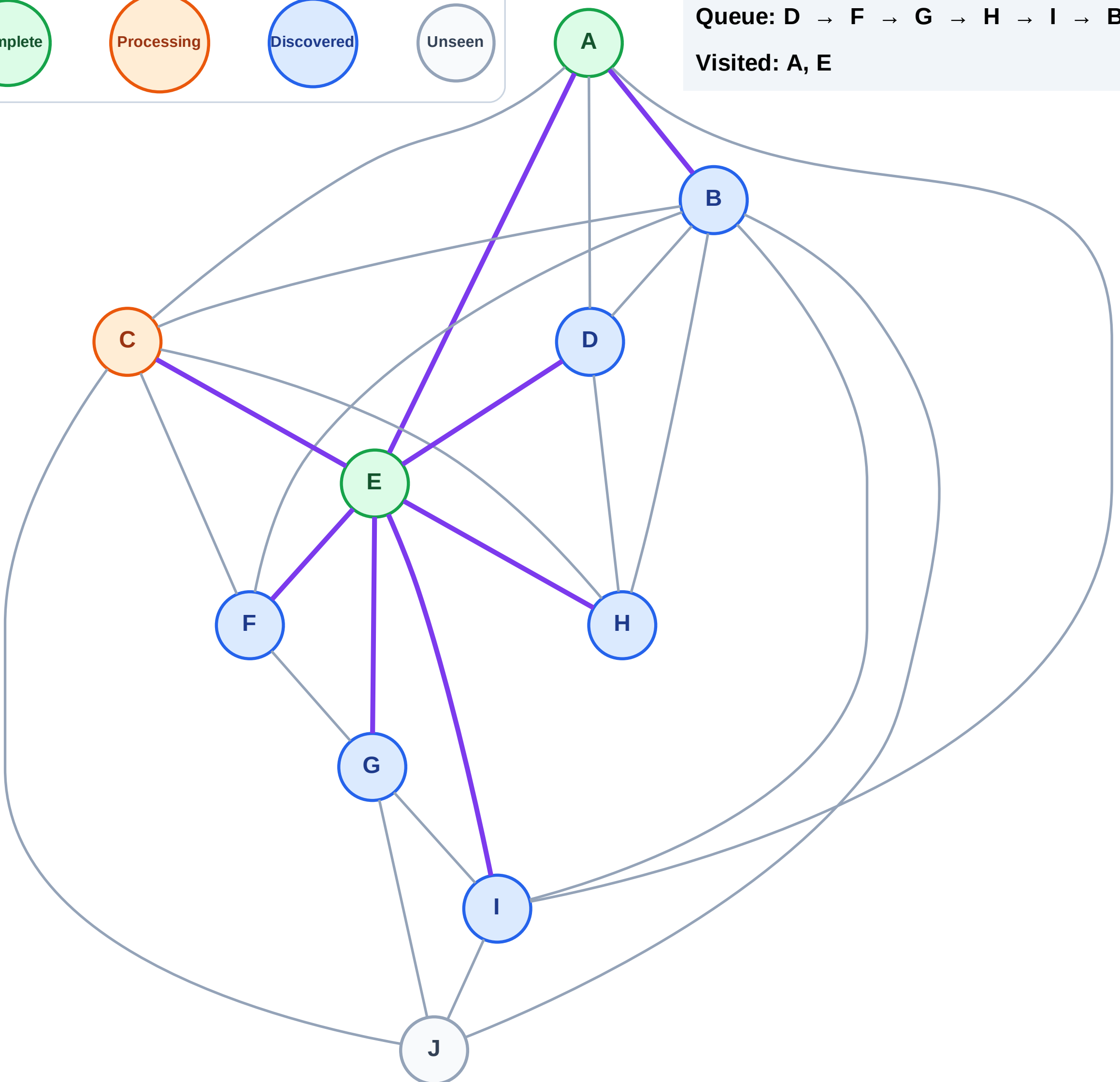
Step 6: Process node C

Legend



Queue: D → F → G → H → I → B

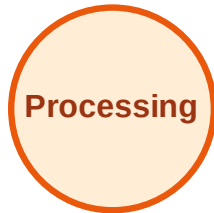
Visited: A, E



BFS Traversal

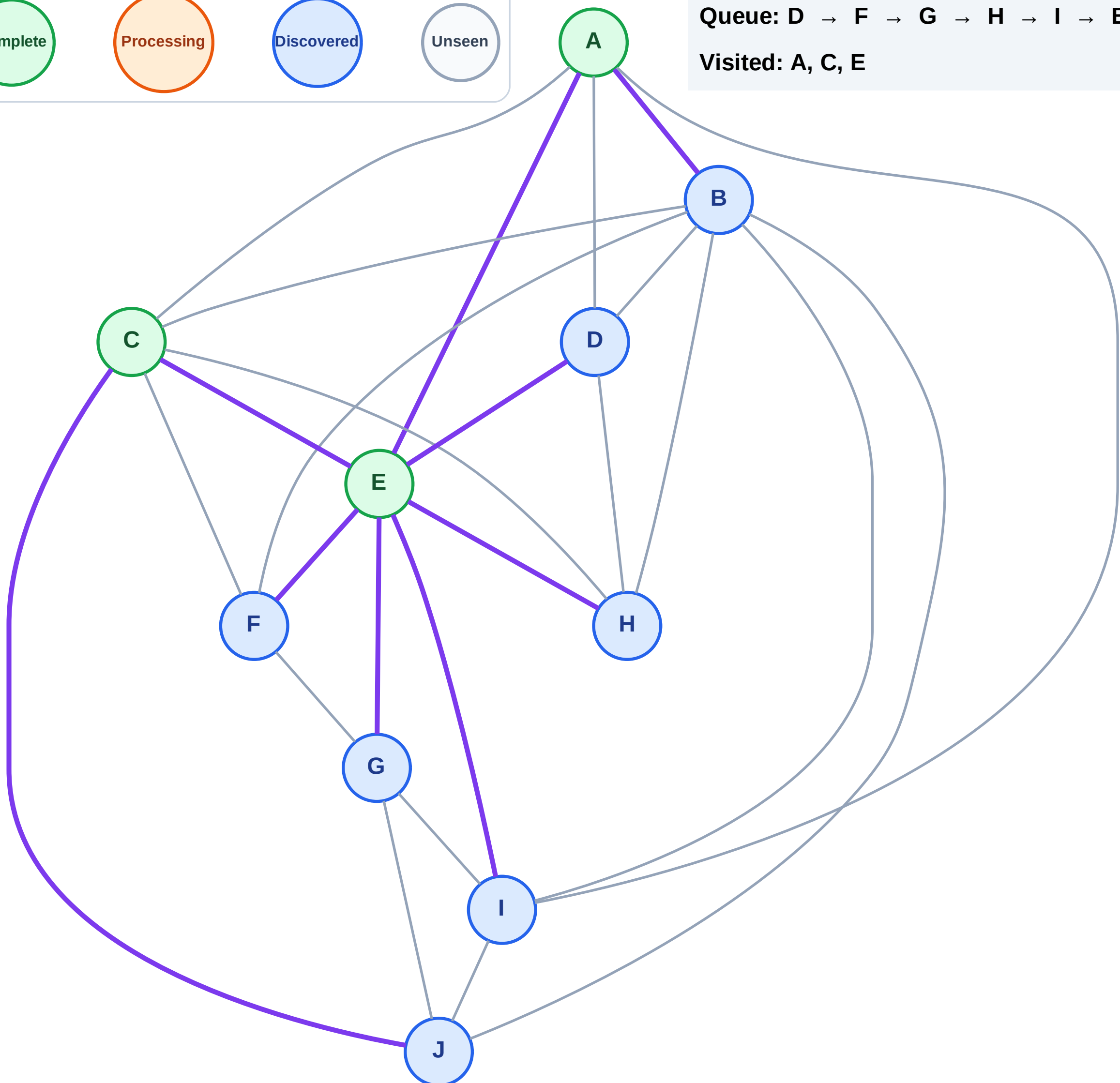
Step 7: Discover J from C

Legend



Queue: D → F → G → H → I → B → J

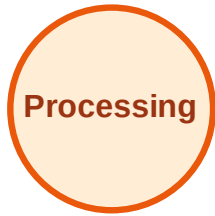
Visited: A, C, E



BFS Traversal

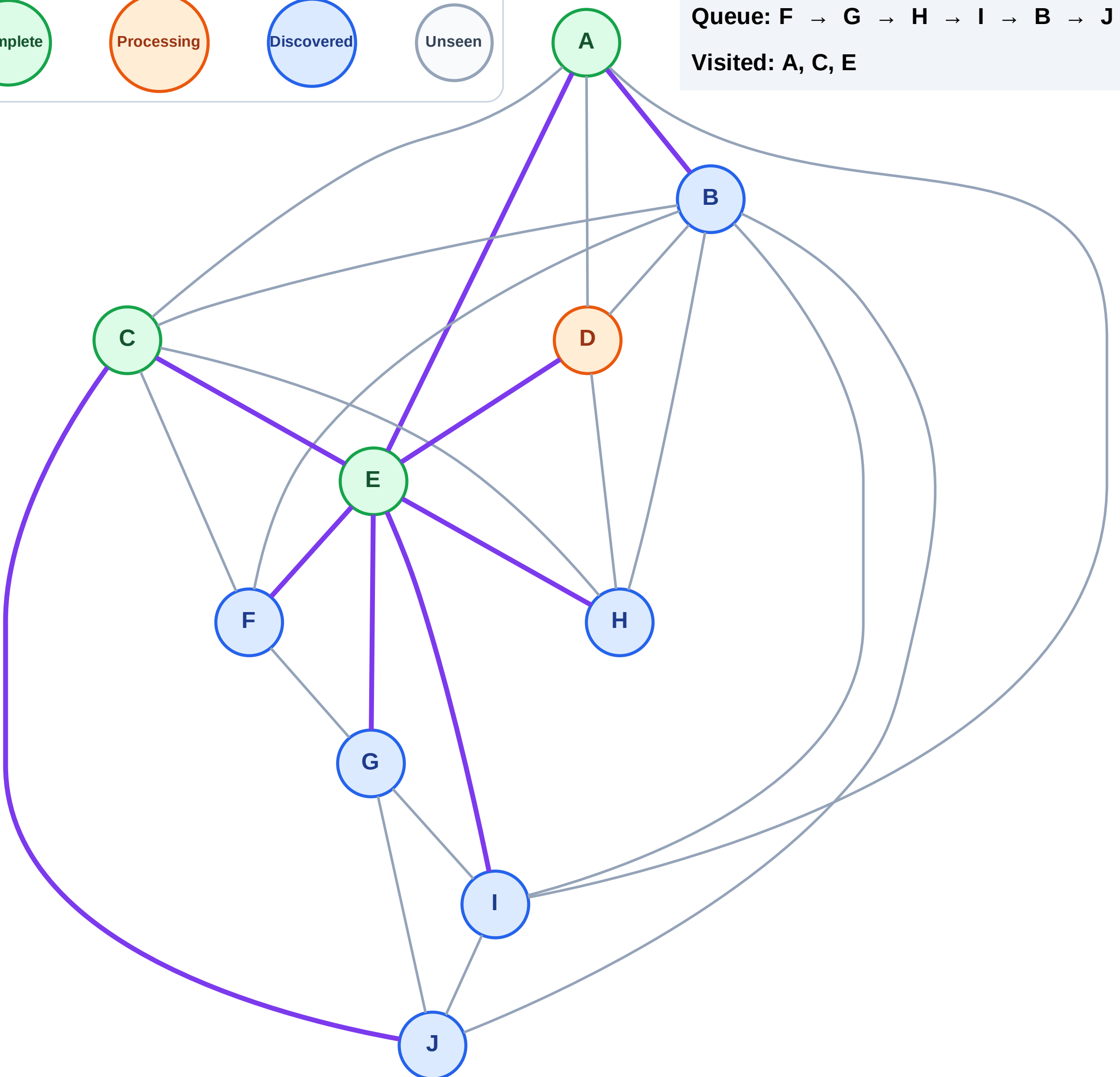
Step 8: Process node D

Legend



Queue: F → G → H → I → B → J

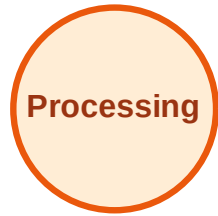
Visited: A, C, E



BFS Traversal

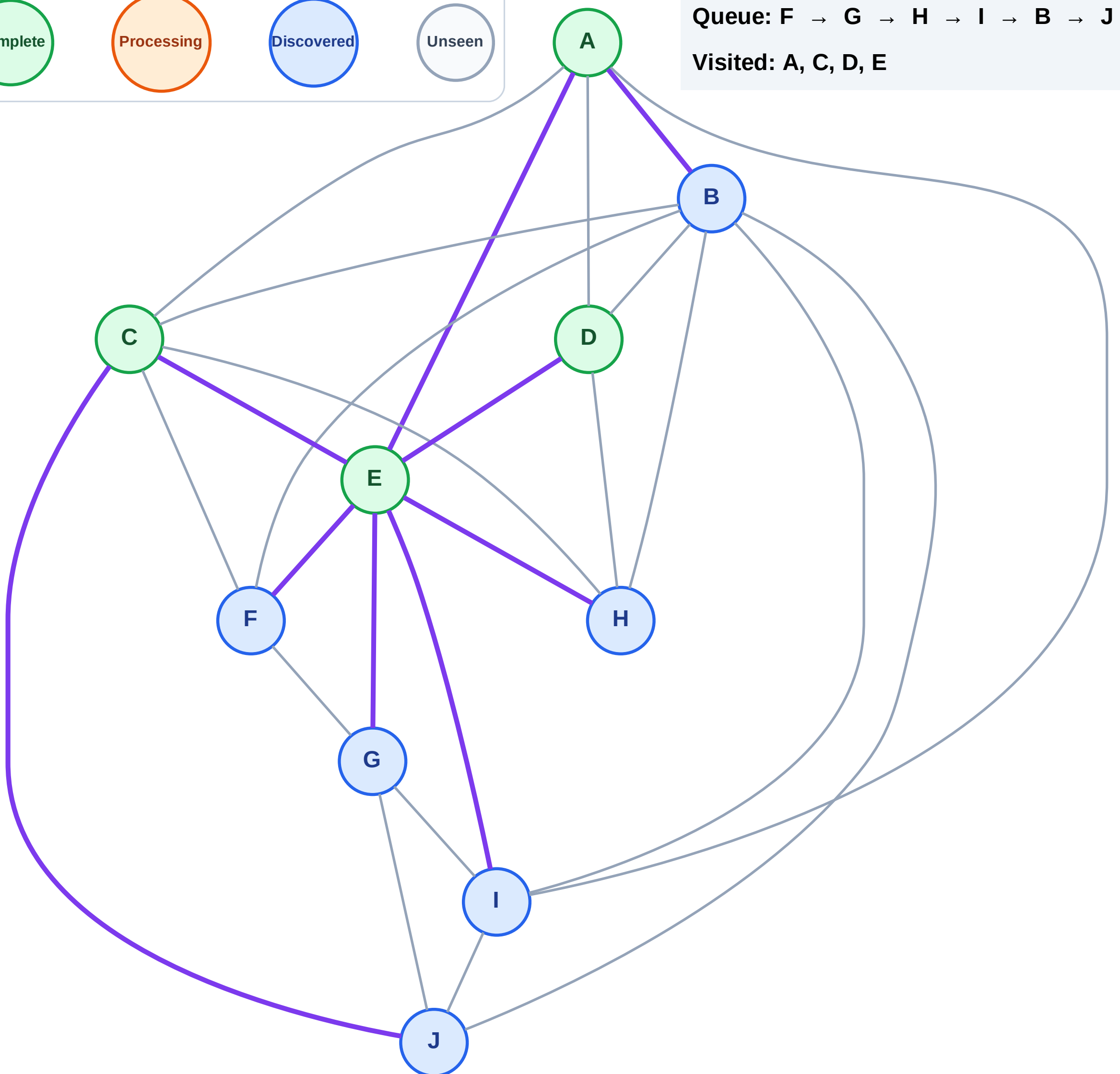
Step 9: No new neighbors from D

Legend



Queue: F → G → H → I → B → J

Visited: A, C, D, E



BFS Traversal

Step 10: Process node F

Complete

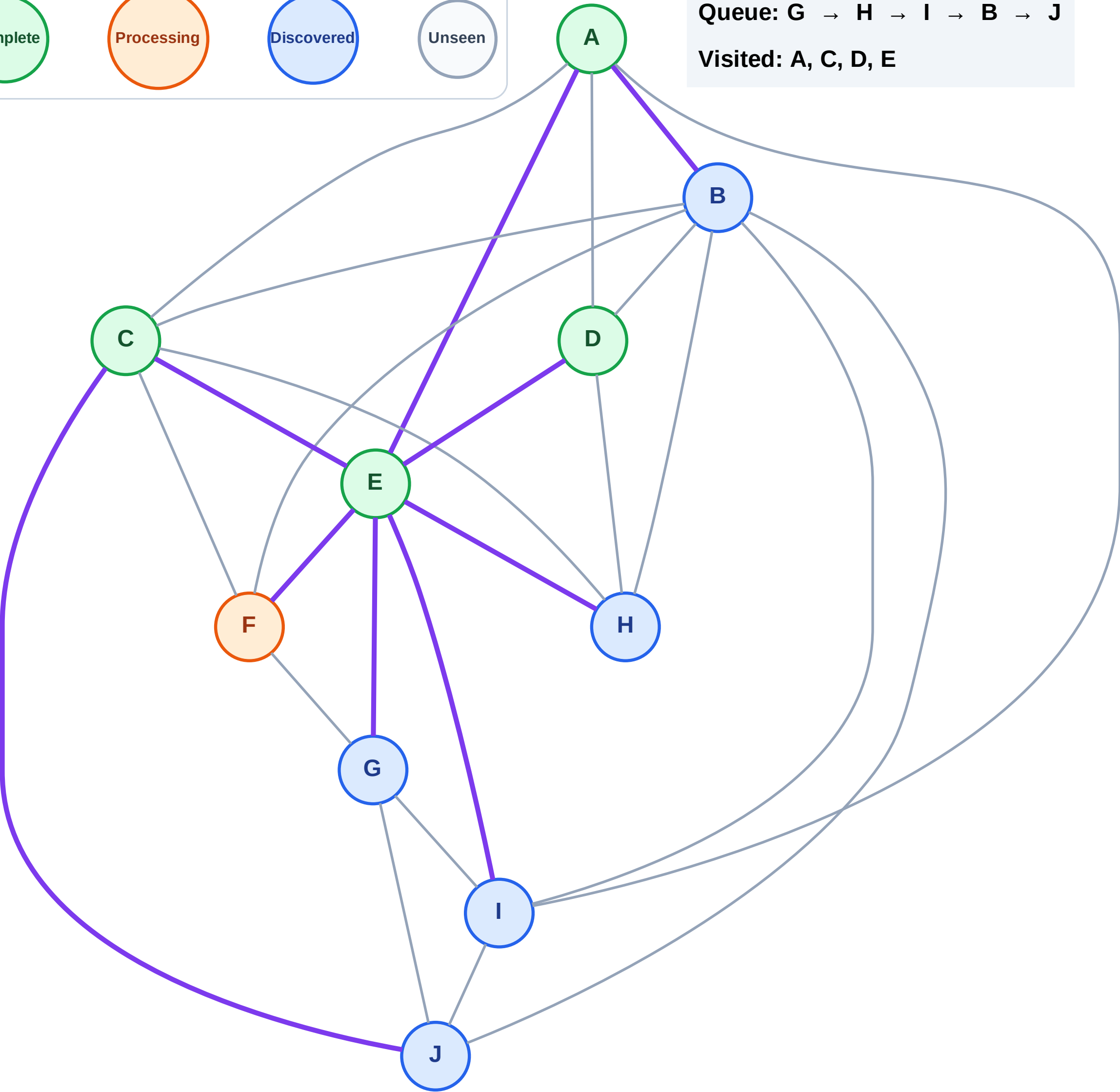
Processing

Discovered

Unseen

Queue: G → H → I → B → J

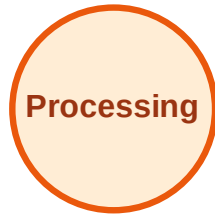
Visited: A, C, D, E



BFS Traversal

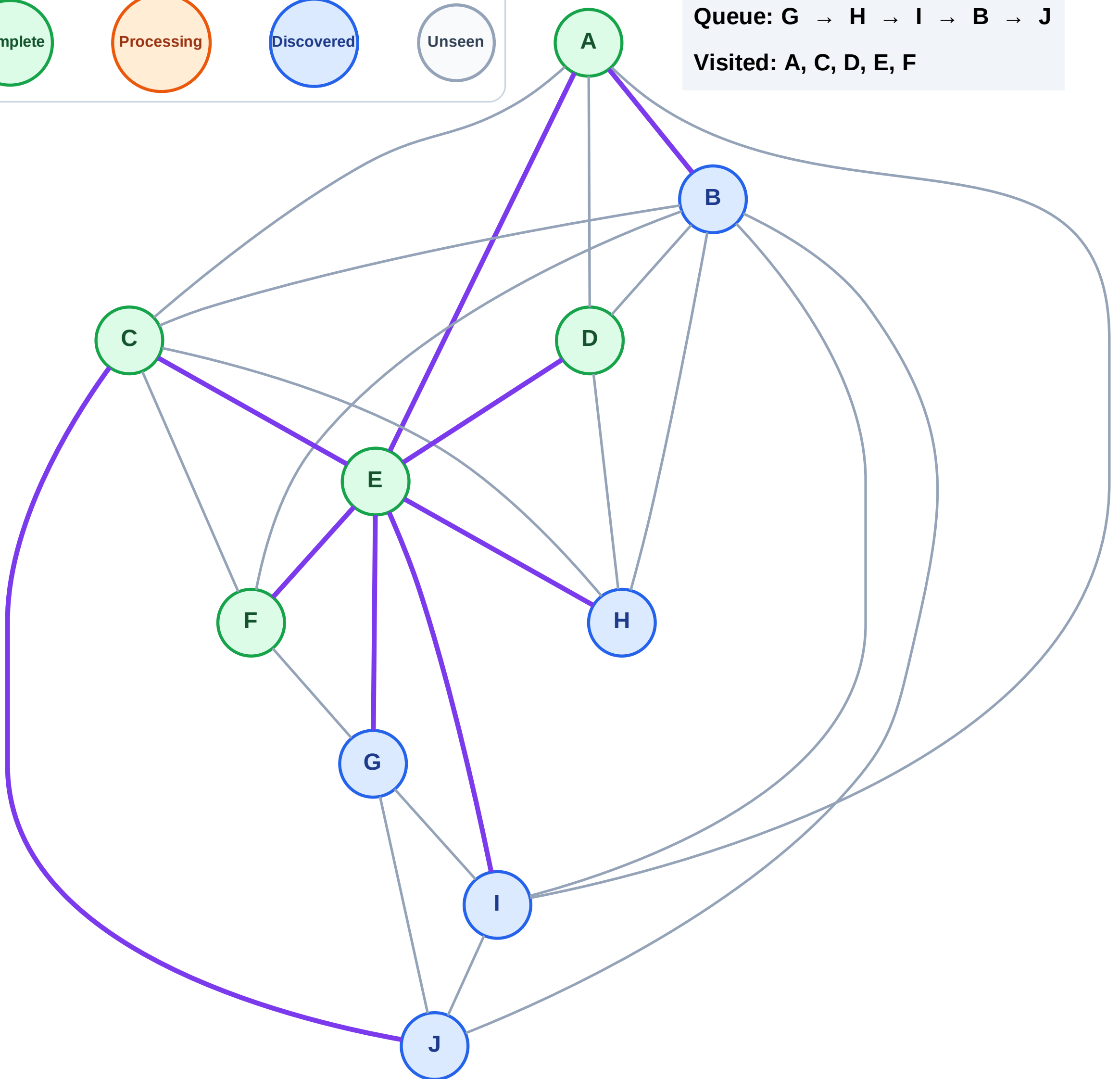
Step 11: No new neighbors from F

Legend



Queue: G → H → I → B → J

Visited: A, C, D, E, F



BFS Traversal

Step 12: Process node G

Complete

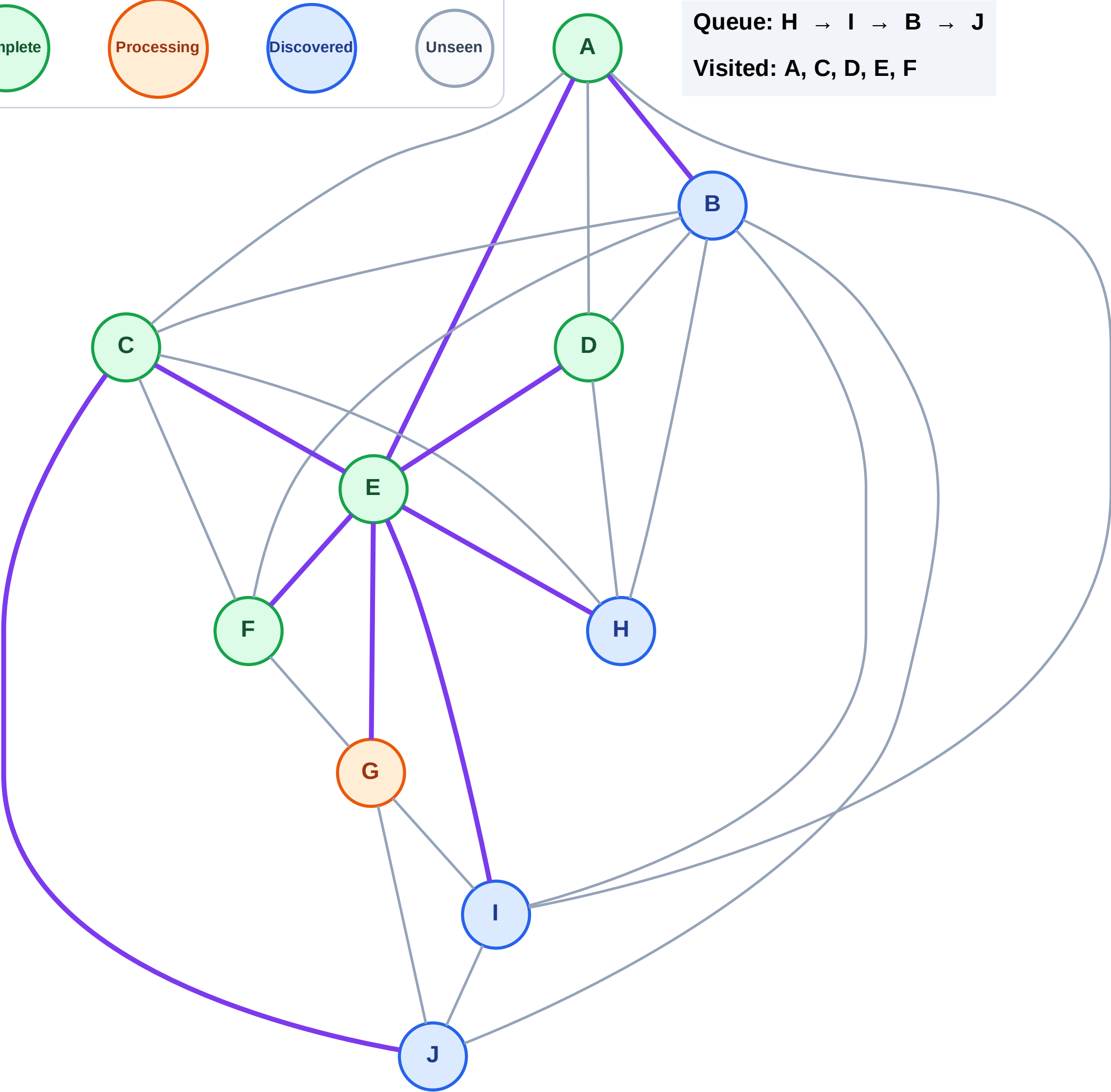
Processing

Discovered

Unseen

Queue: H → I → B → J

Visited: A, C, D, E, F



BFS Traversal

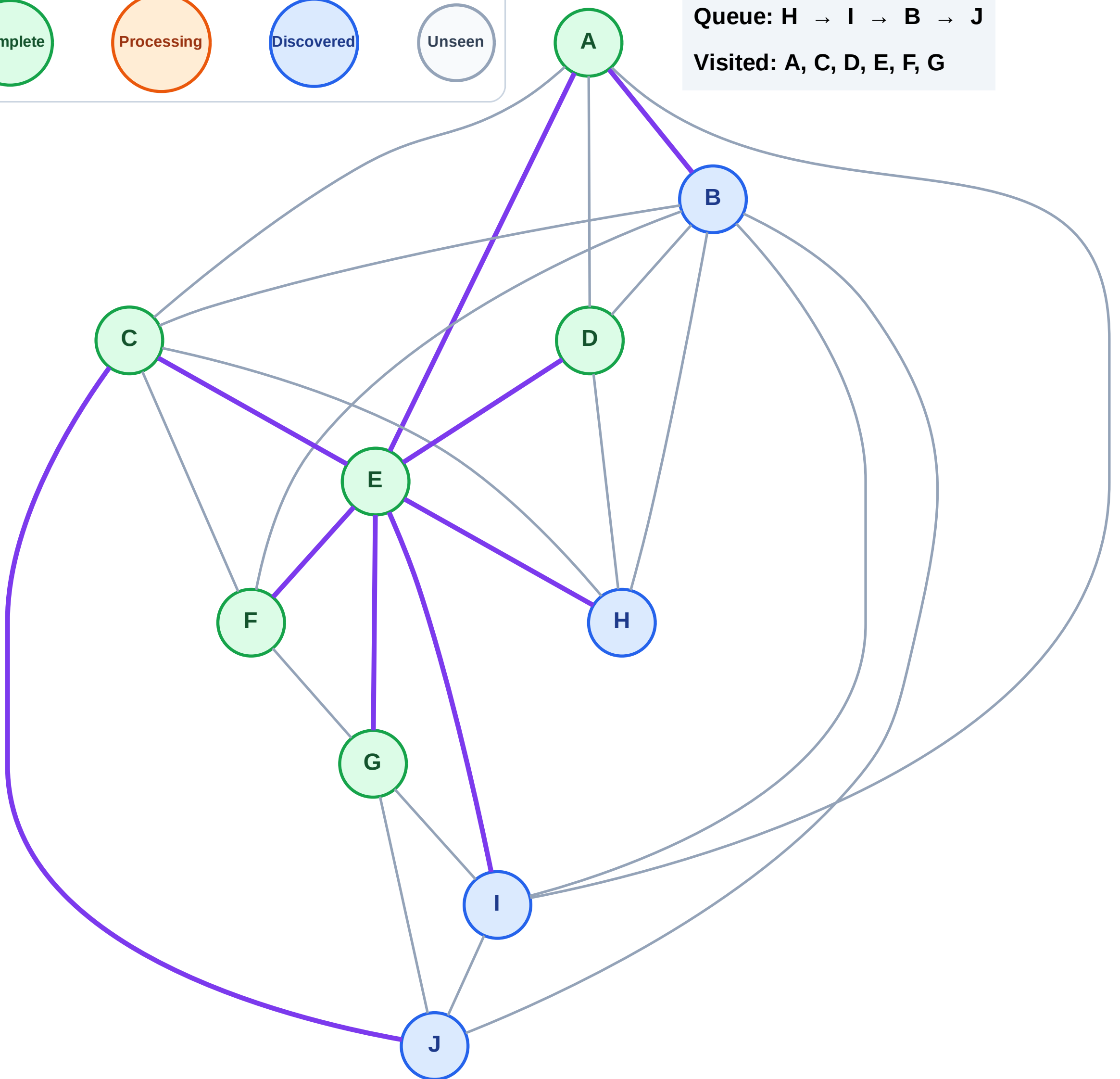
Step 13: No new neighbors from G

Legend



Queue: H → I → B → J

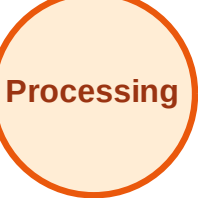
Visited: A, C, D, E, F, G



BFS Traversal

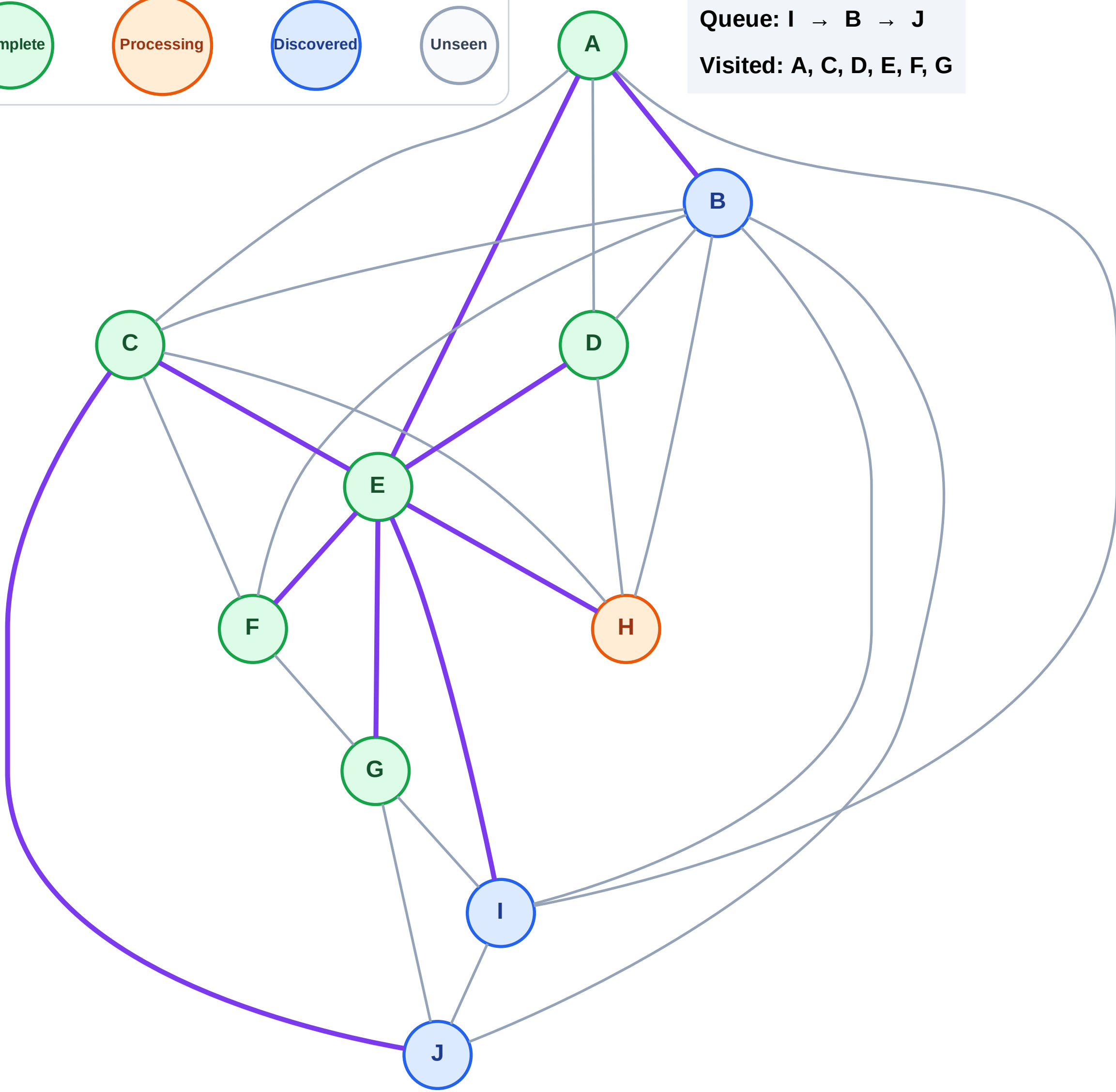
Step 14: Process node H

Legend



Queue: I → B → J

Visited: A, C, D, E, F, G



BFS Traversal

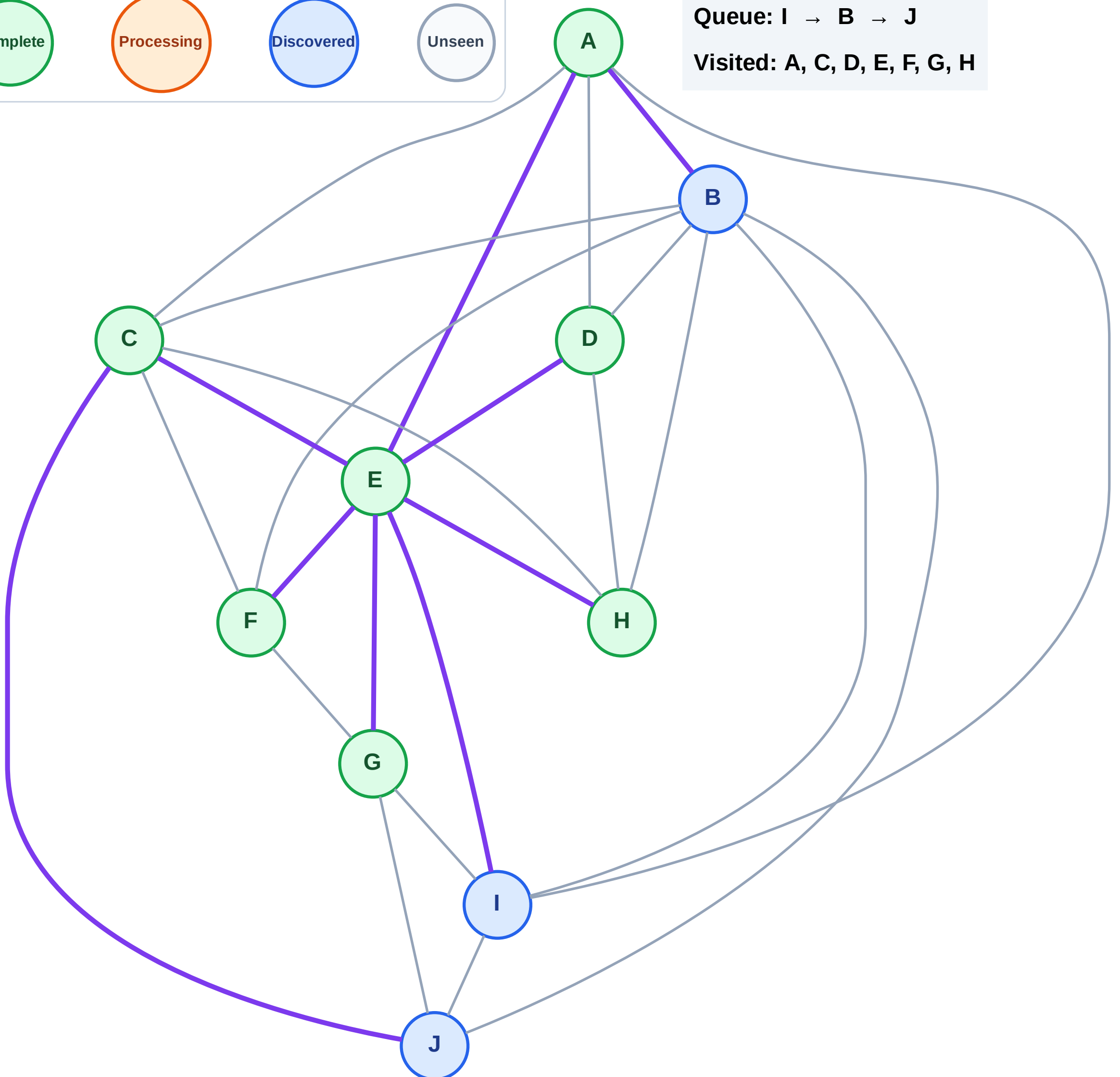
Step 15: No new neighbors from H

Legend



Queue: I → B → J

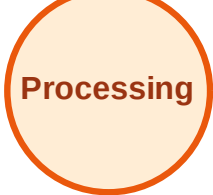
Visited: A, C, D, E, F, G, H



BFS Traversal

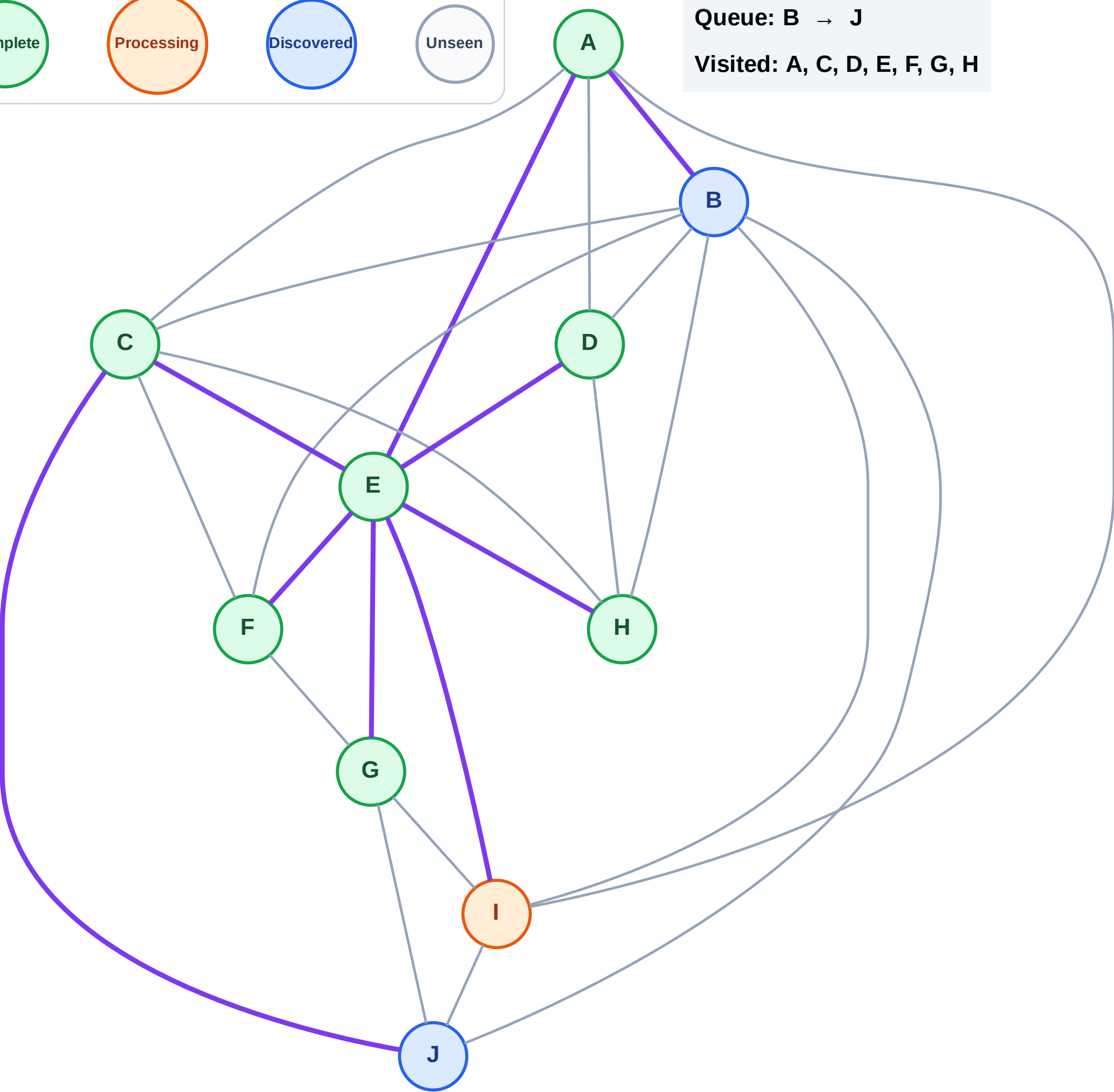
Step 16: Process node I

Legend



Queue: B → J

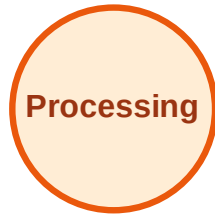
Visited: A, C, D, E, F, G, H



BFS Traversal

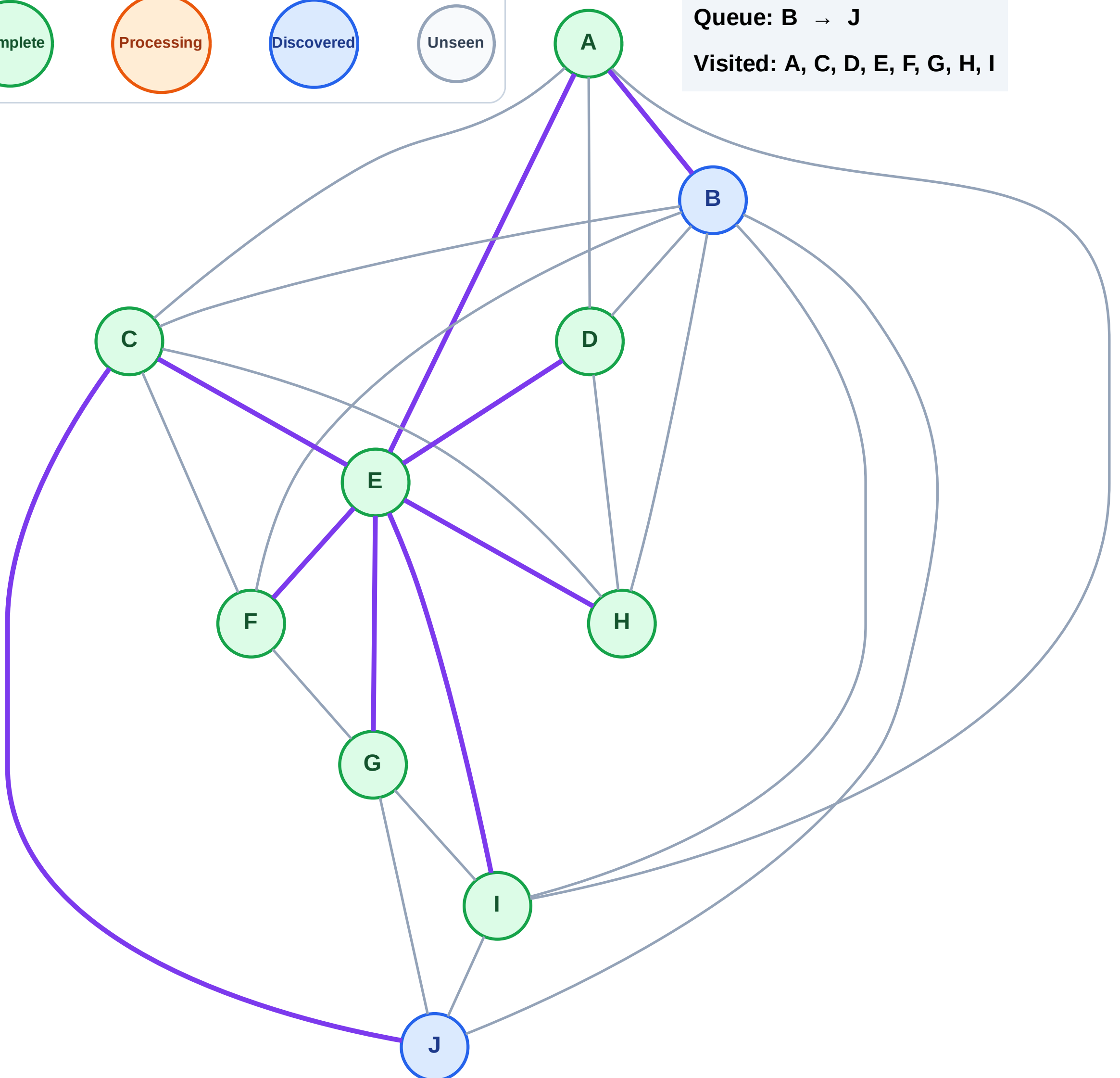
Step 17: No new neighbors from I

Legend



Queue: B → J

Visited: A, C, D, E, F, G, H, I



BFS Traversal

Step 18: Process node B

Legend

Complete

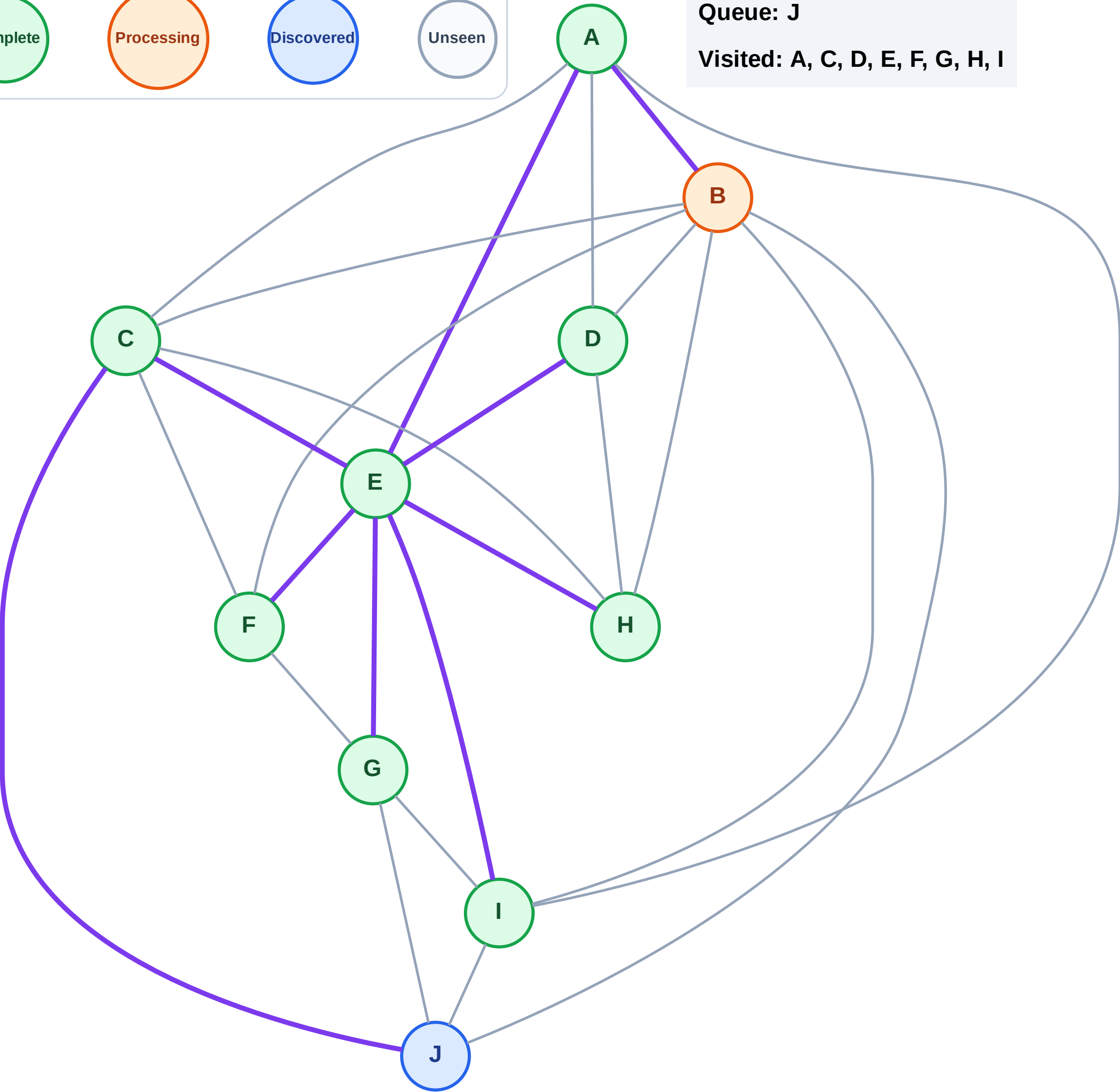
Processing

Discovered

Unseen

Queue: J

Visited: A, C, D, E, F, G, H, I



BFS Traversal

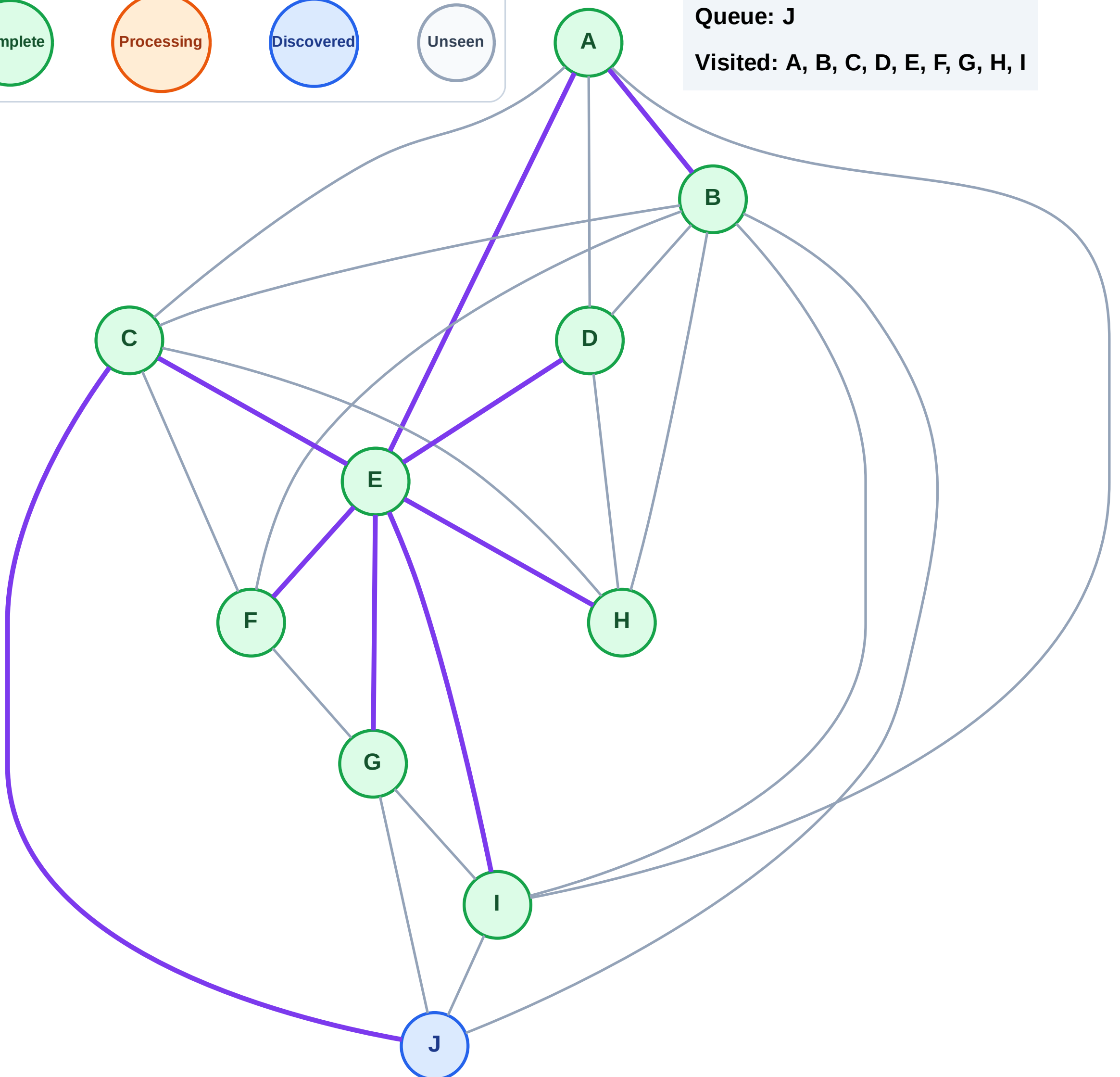
Step 19: No new neighbors from B

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

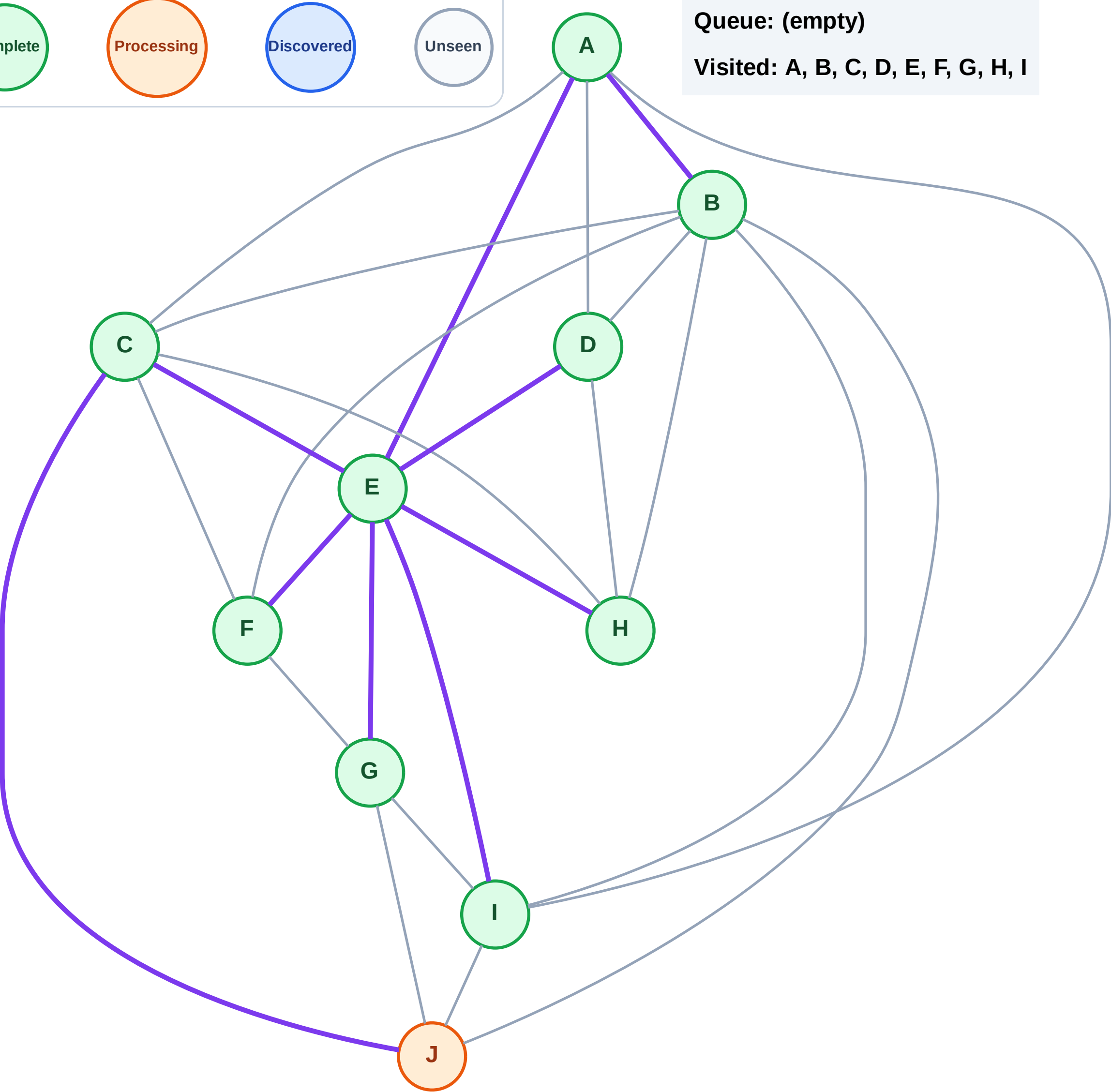
Step 20: Process node J

Legend



Queue: (empty)

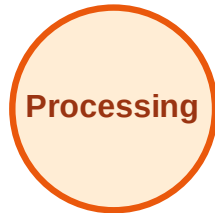
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

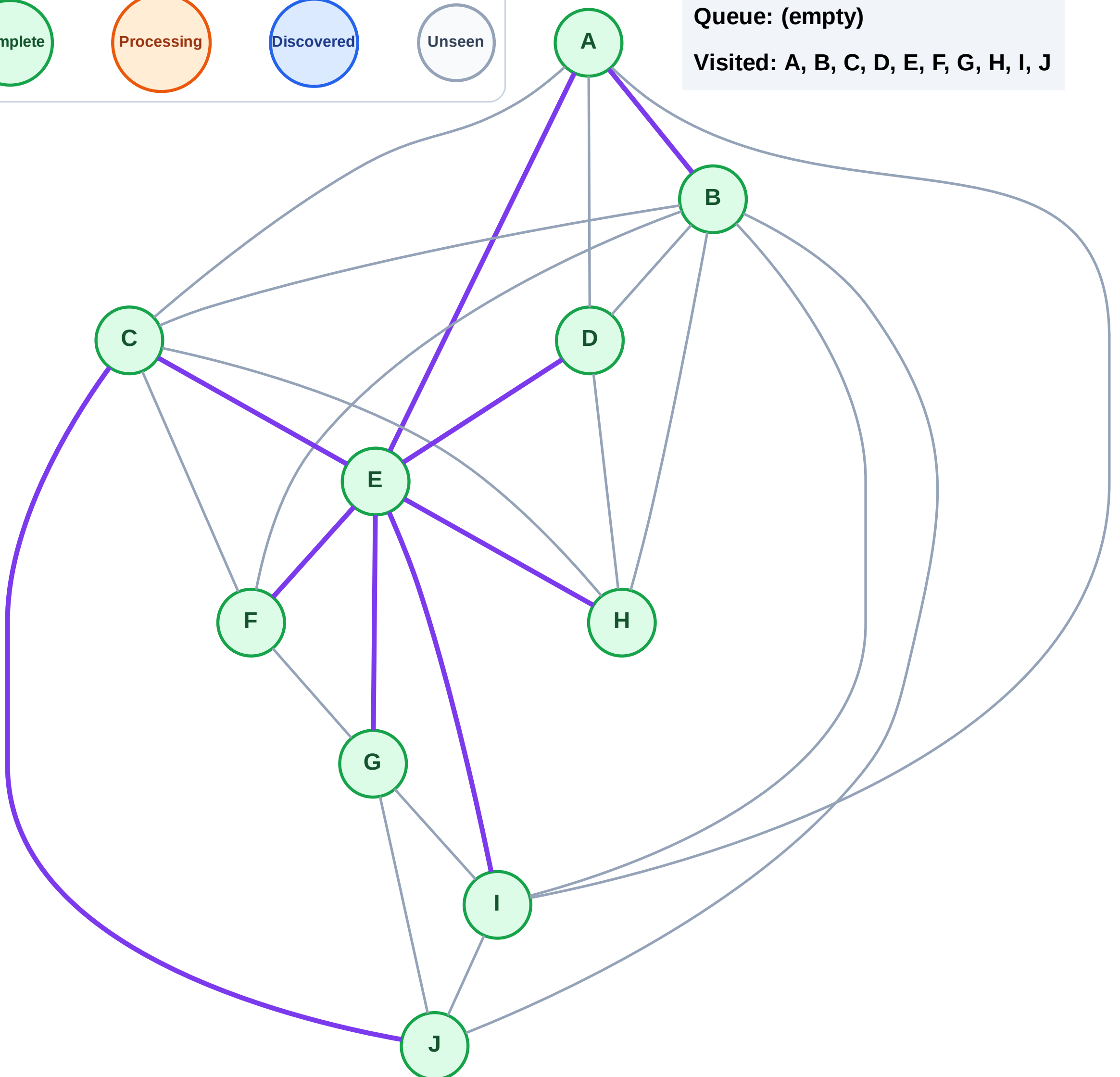
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

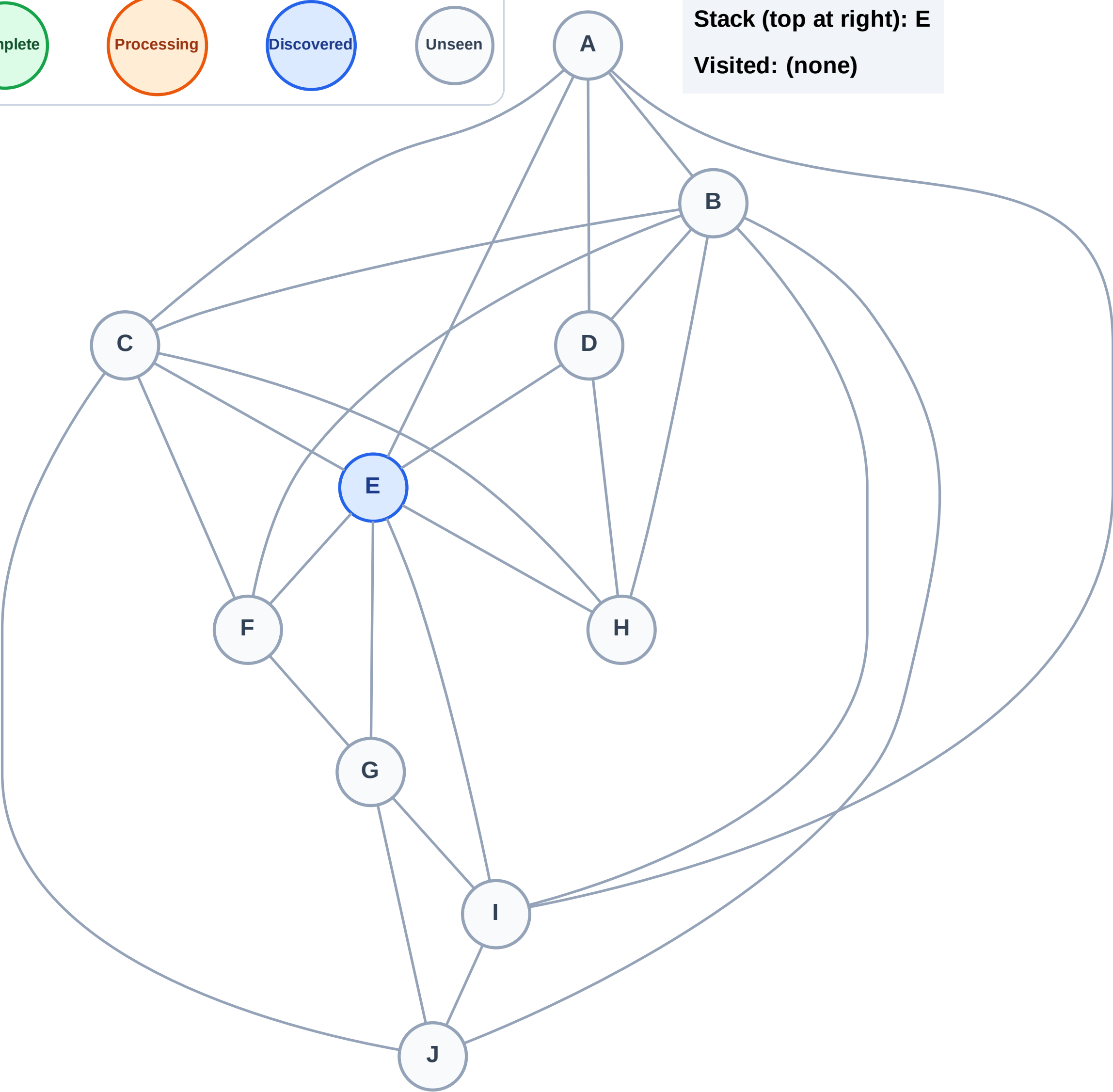
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

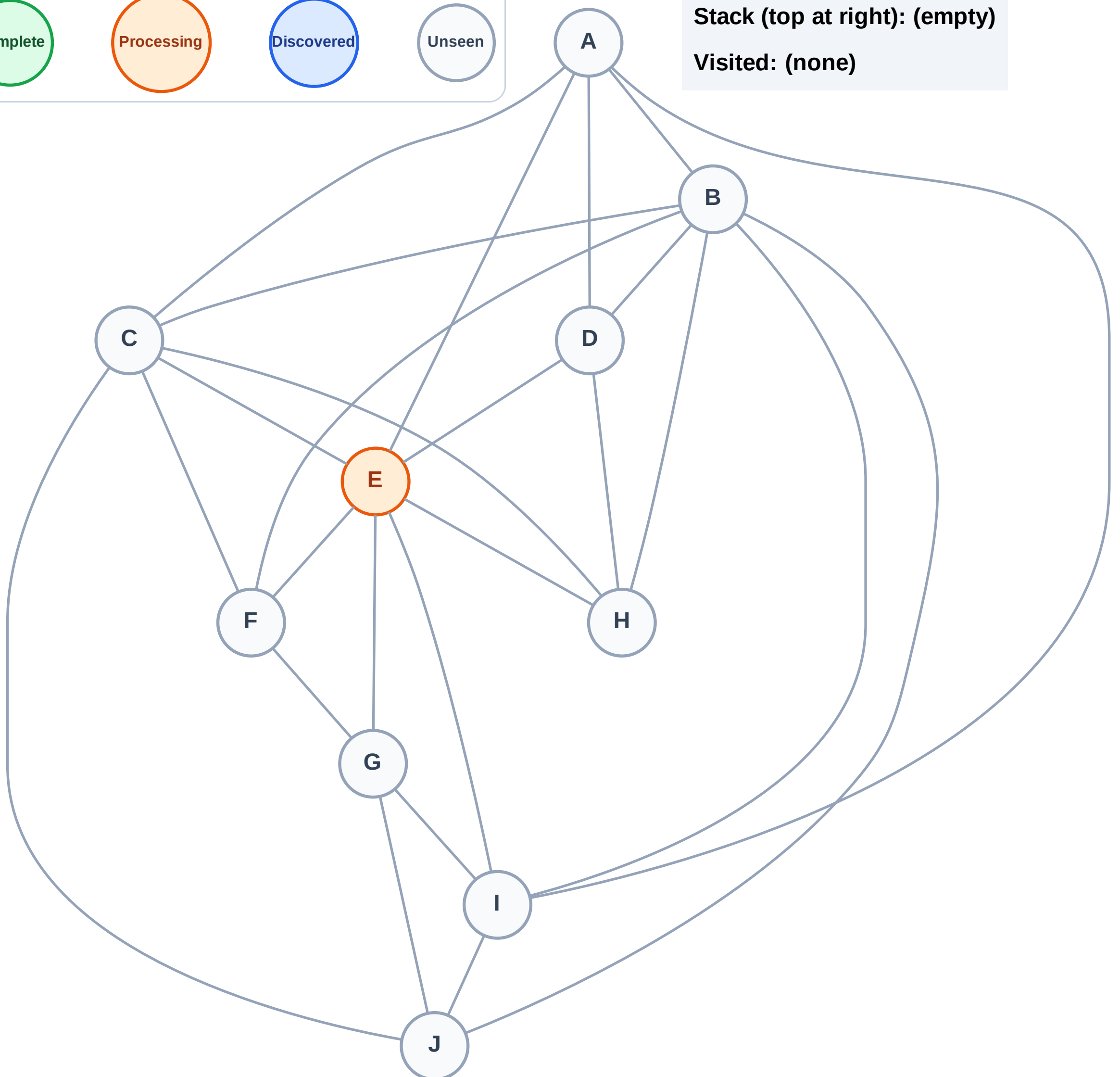
Step 2: Process node E

Legend



Stack (top at right): (empty)

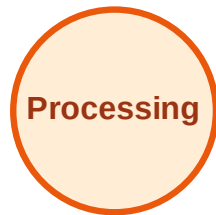
Visited: (none)



DFS Traversal

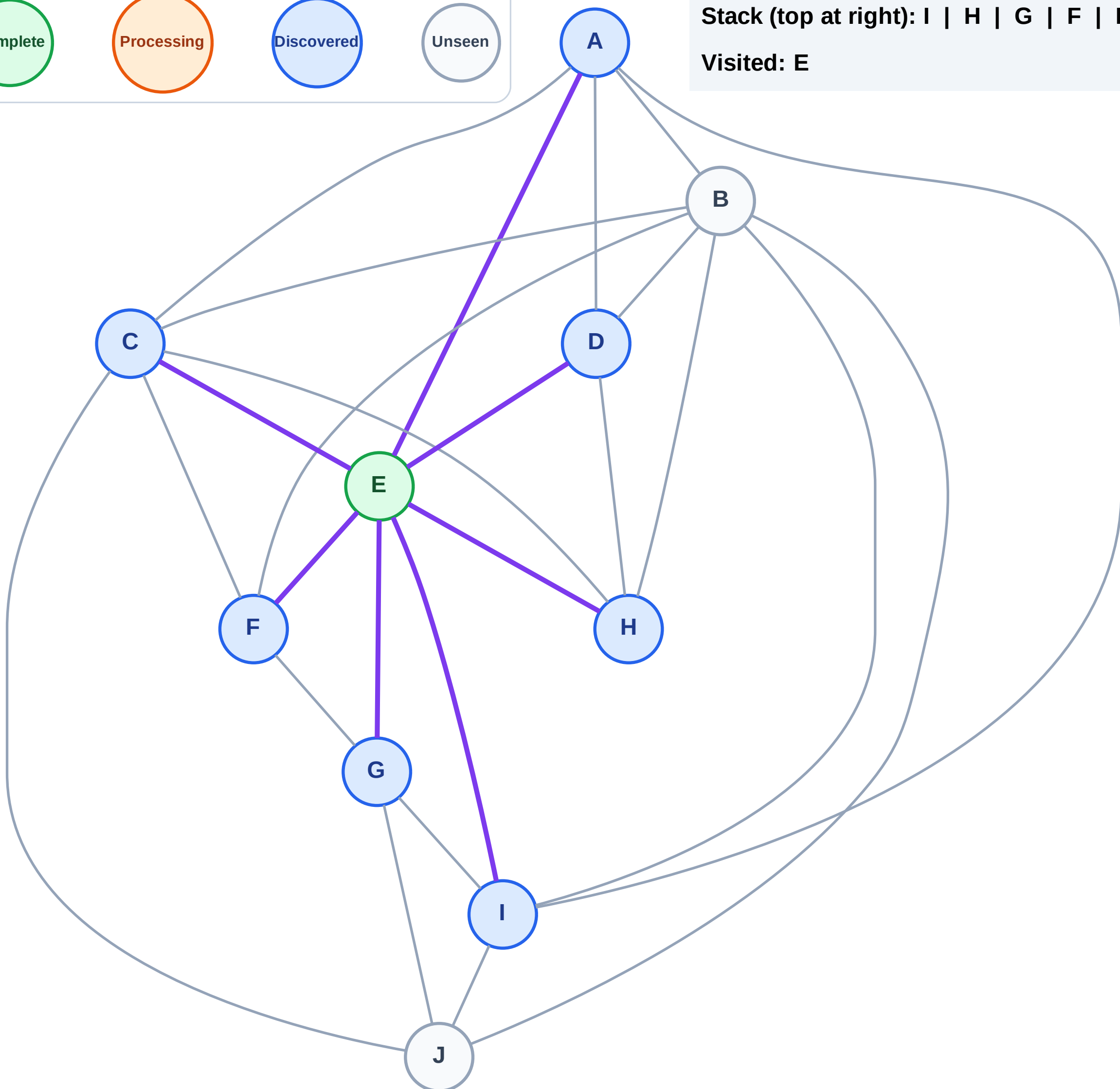
Step 3: Discover I, H, G, F, D, C, A from E

Legend



Stack (top at right): I | H | G | F | D | C | A

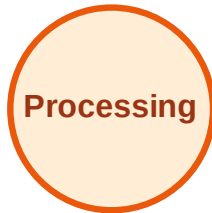
Visited: E



DFS Traversal

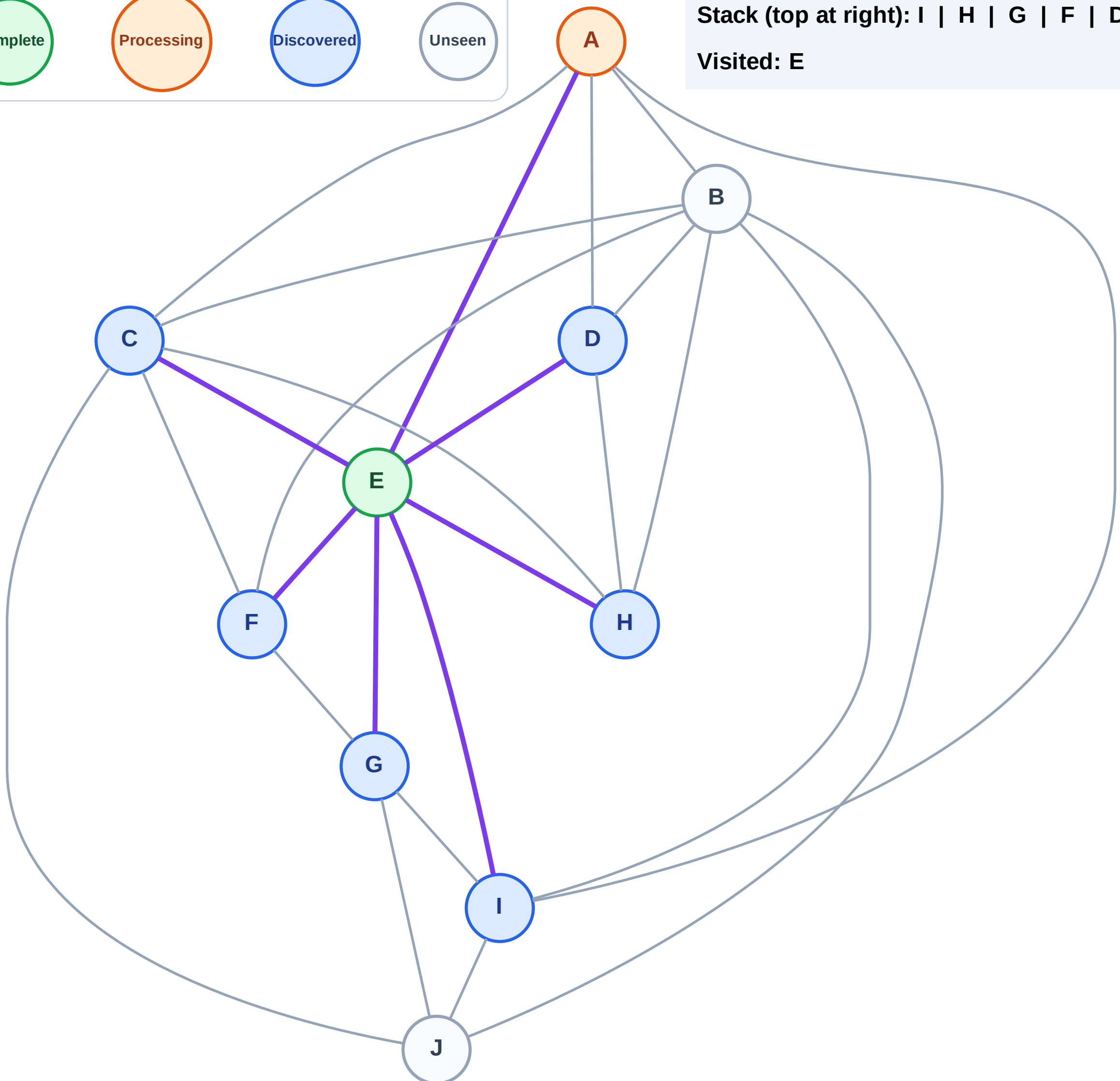
Step 4: Process node A

Legend



Stack (top at right): I | H | G | F | D | C

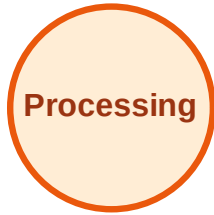
Visited: E



DFS Traversal

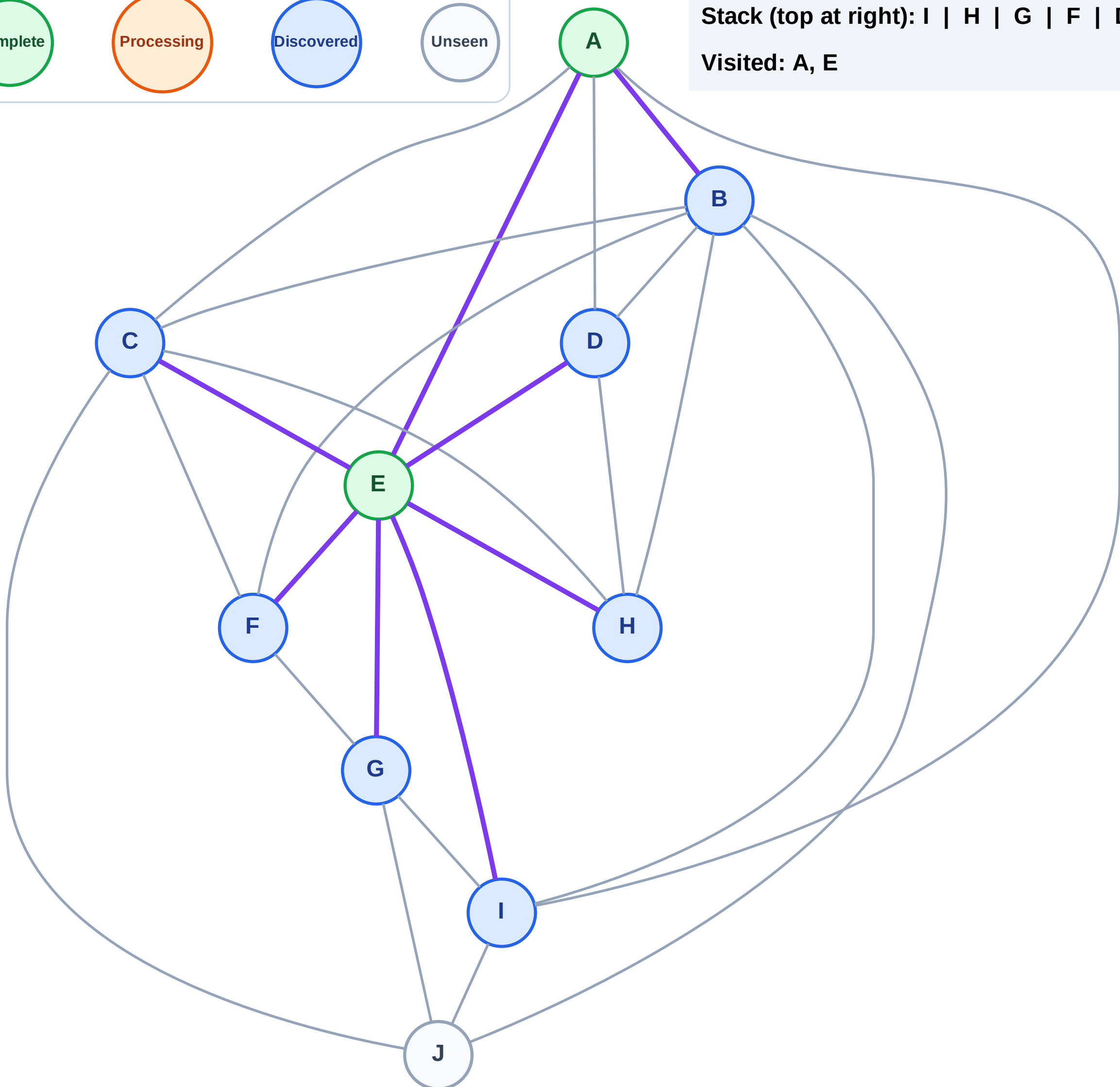
Step 5: Discover B from A

Legend



Stack (top at right): I | H | G | F | D | C | B

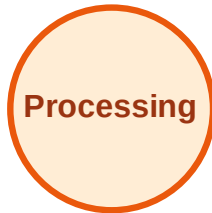
Visited: A, E



DFS Traversal

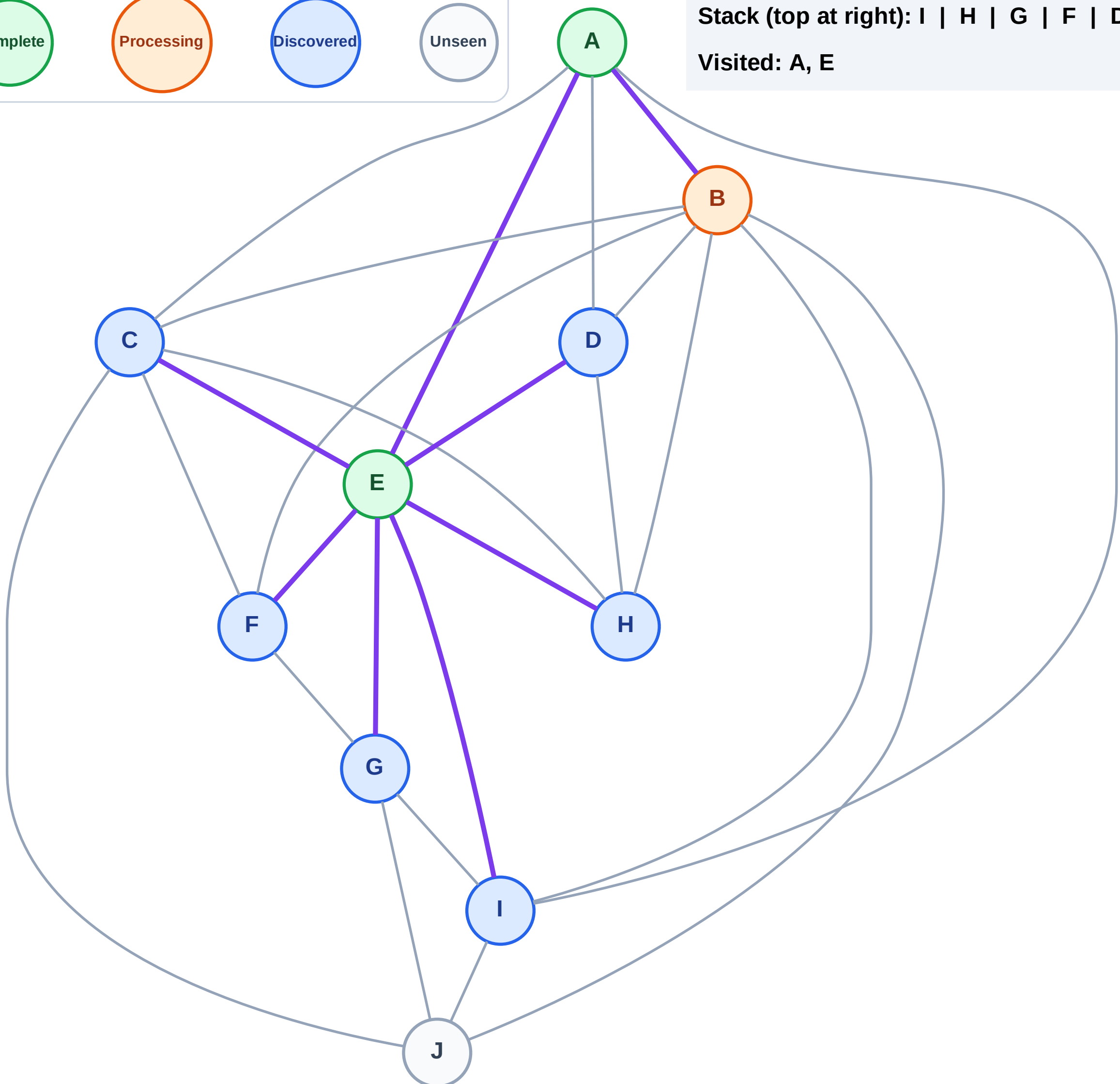
Step 6: Process node B

Legend



Stack (top at right): I | H | G | F | D | C

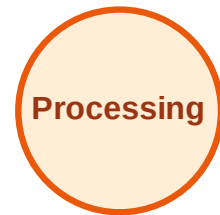
Visited: A, E



DFS Traversal

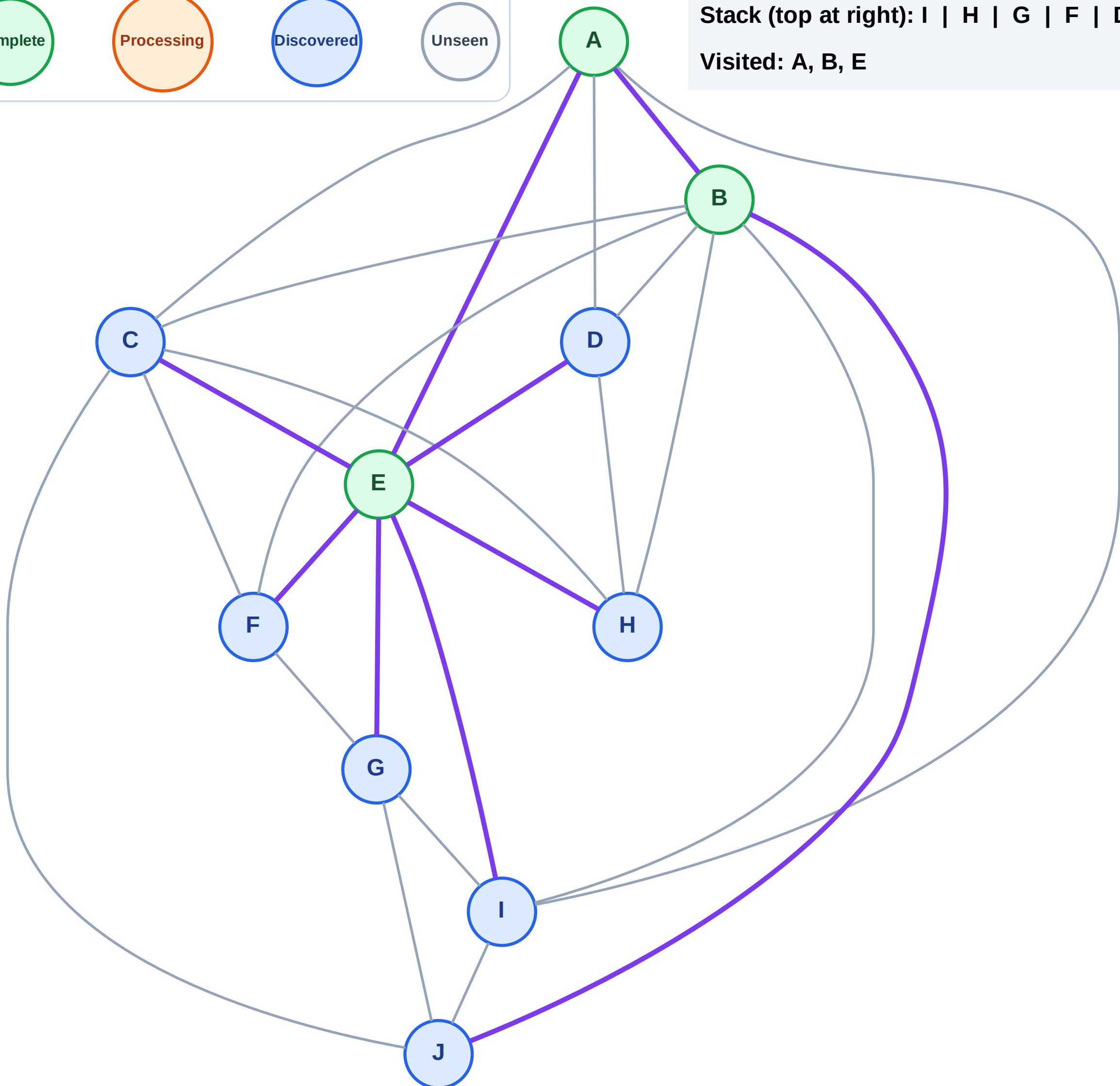
Step 7: Discover J from B

Legend



Stack (top at right): I | H | G | F | D | C | J

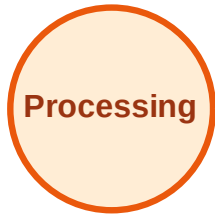
Visited: A, B, E



DFS Traversal

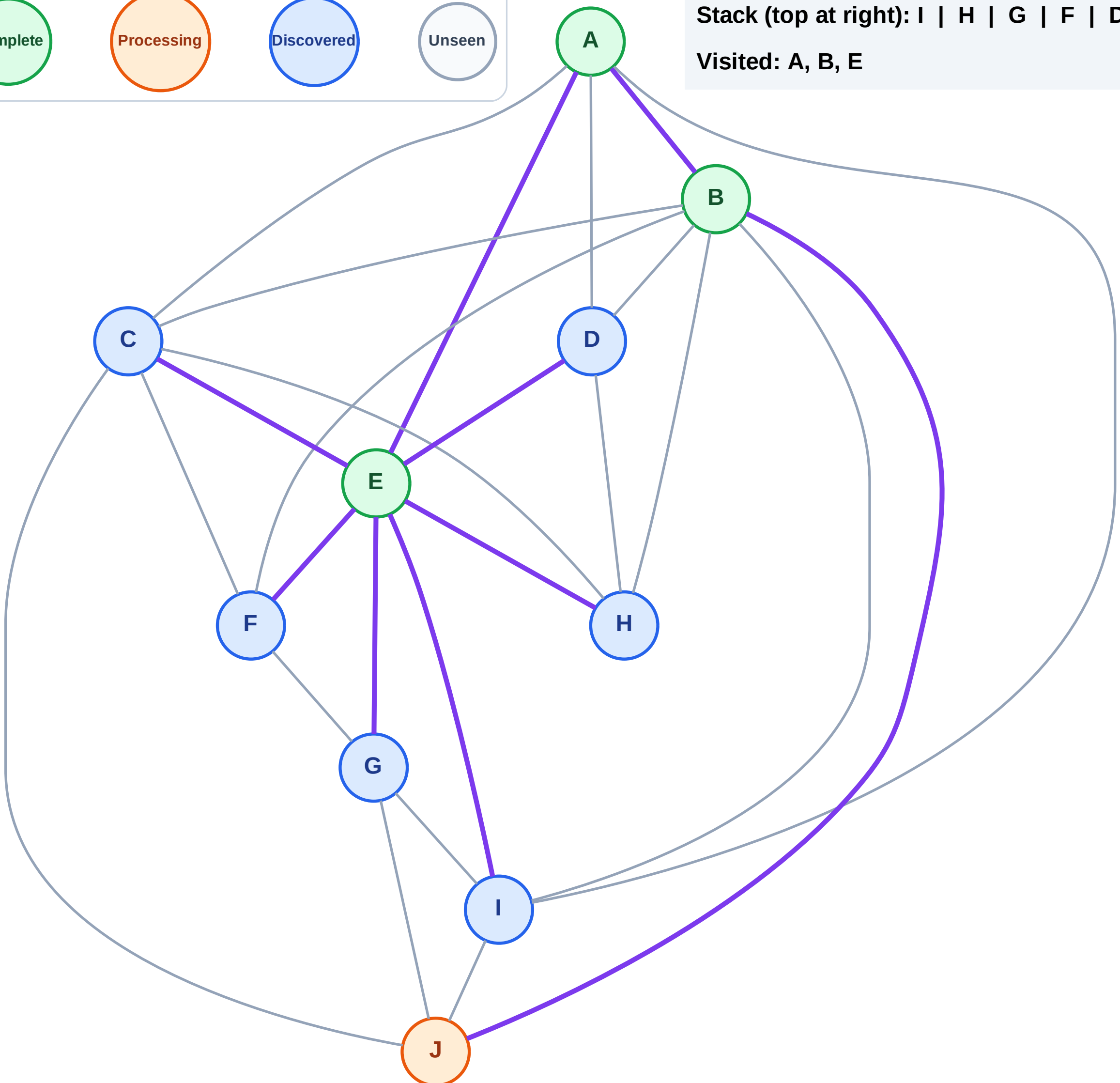
Step 8: Process node J

Legend



Stack (top at right): I | H | G | F | D | C

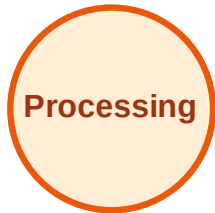
Visited: A, B, E



DFS Traversal

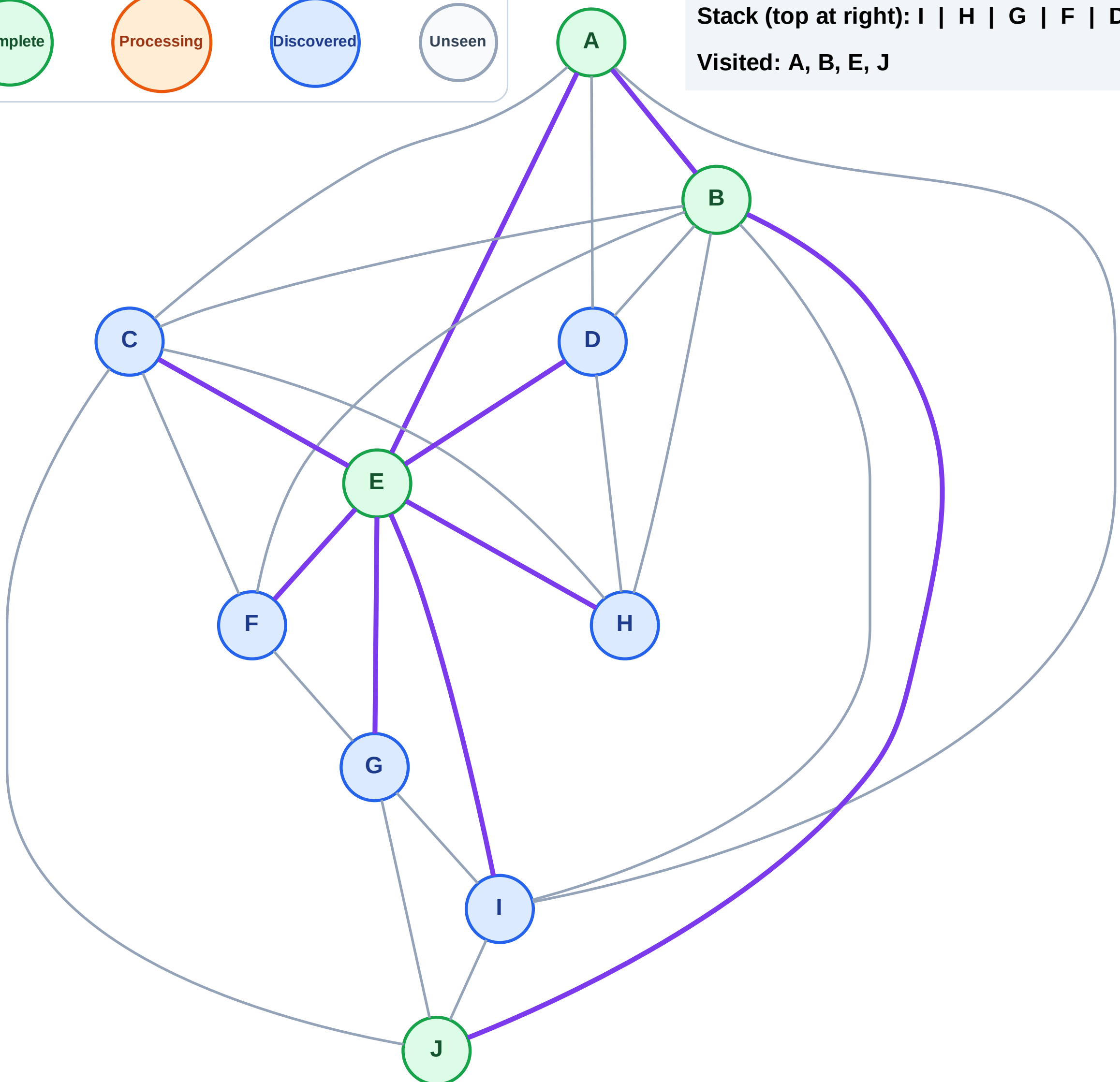
Step 9: No new neighbors from J

Legend



Stack (top at right): I | H | G | F | D | C

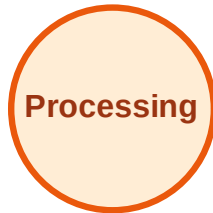
Visited: A, B, E, J



DFS Traversal

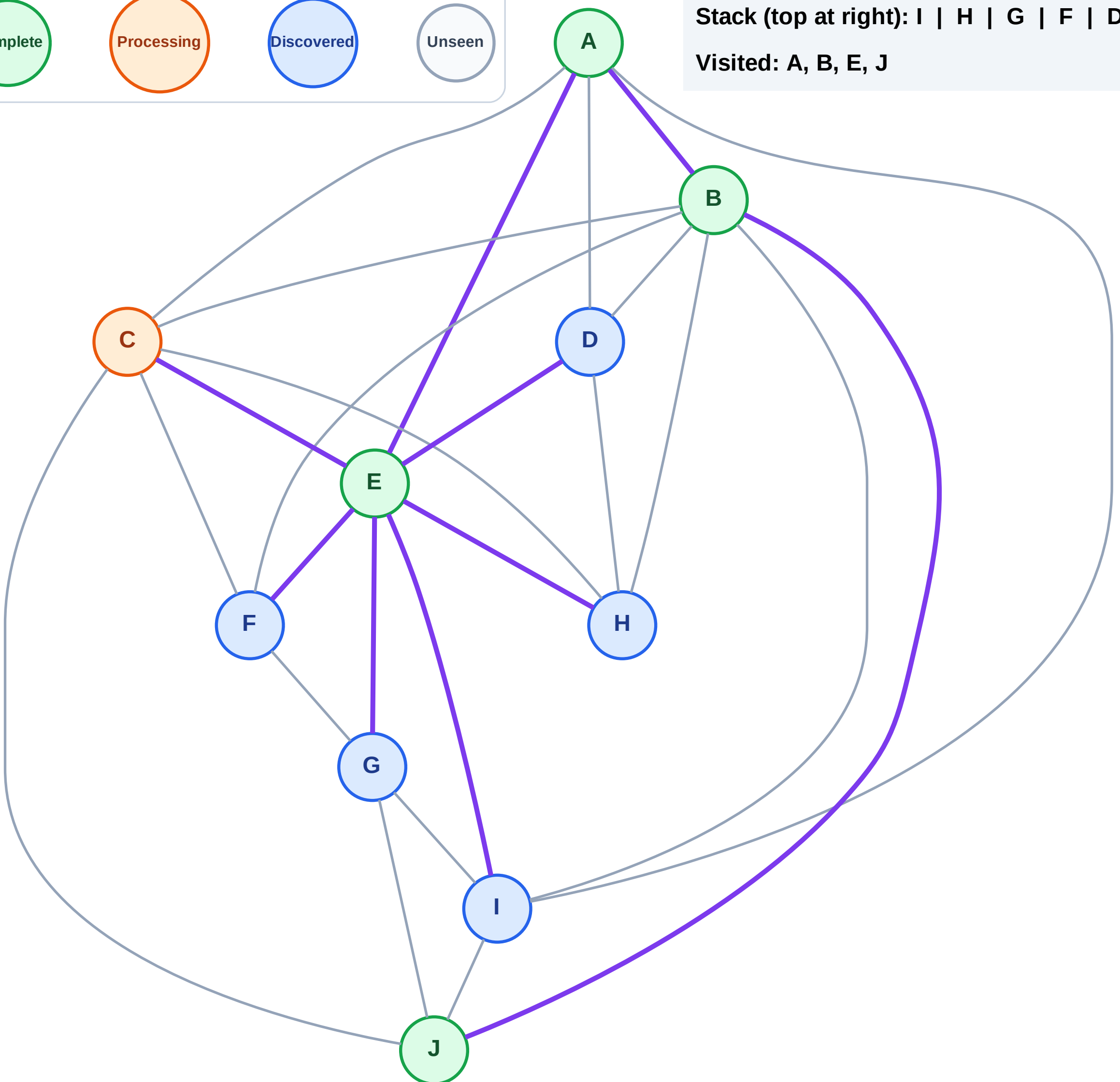
Step 10: Process node C

Legend



Stack (top at right): I | H | G | F | D

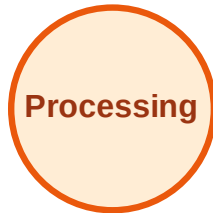
Visited: A, B, E, J



DFS Traversal

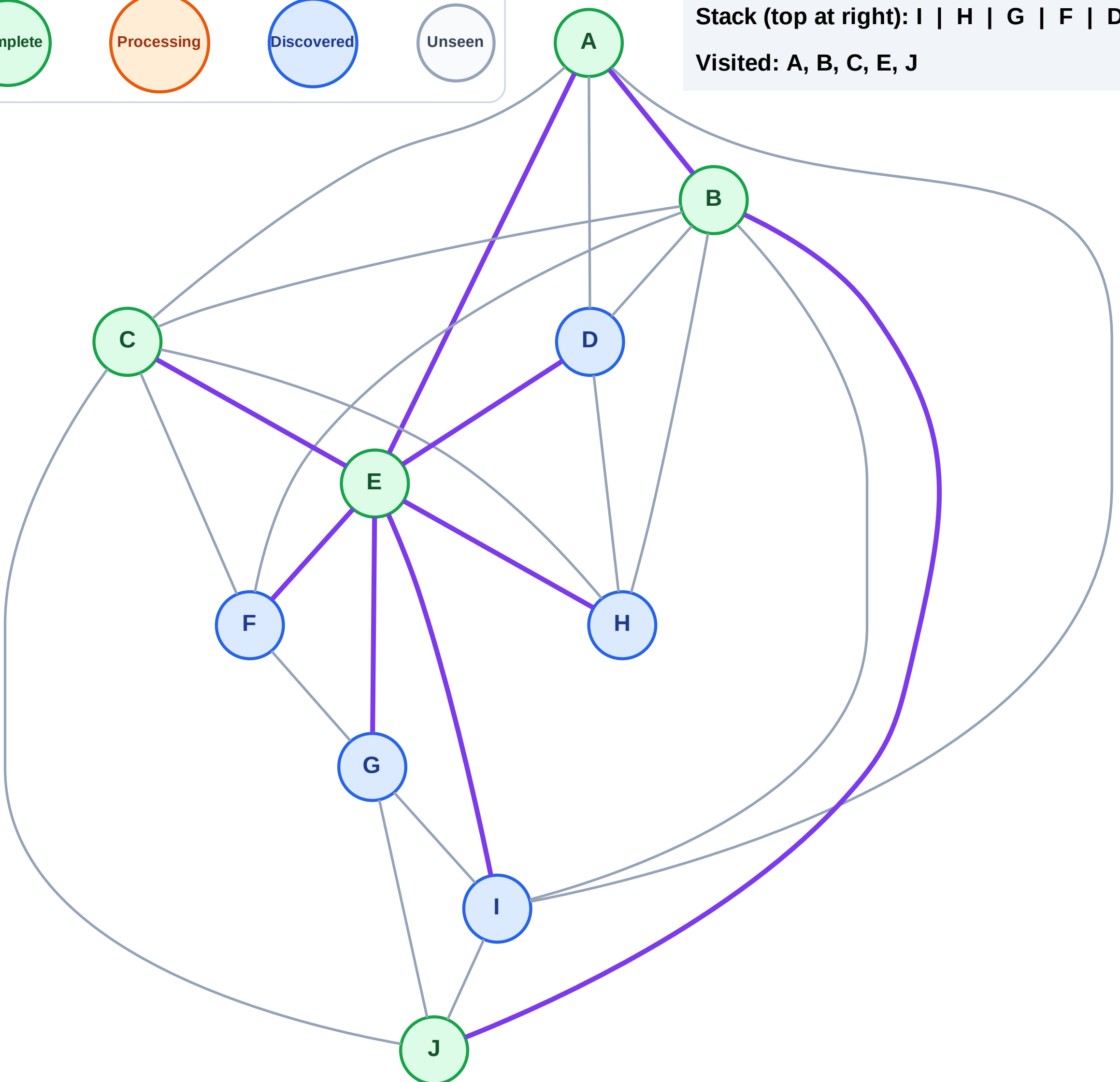
Step 11: No new neighbors from C

Legend



Stack (top at right): I | H | G | F | D

Visited: A, B, C, E, J



DFS Traversal

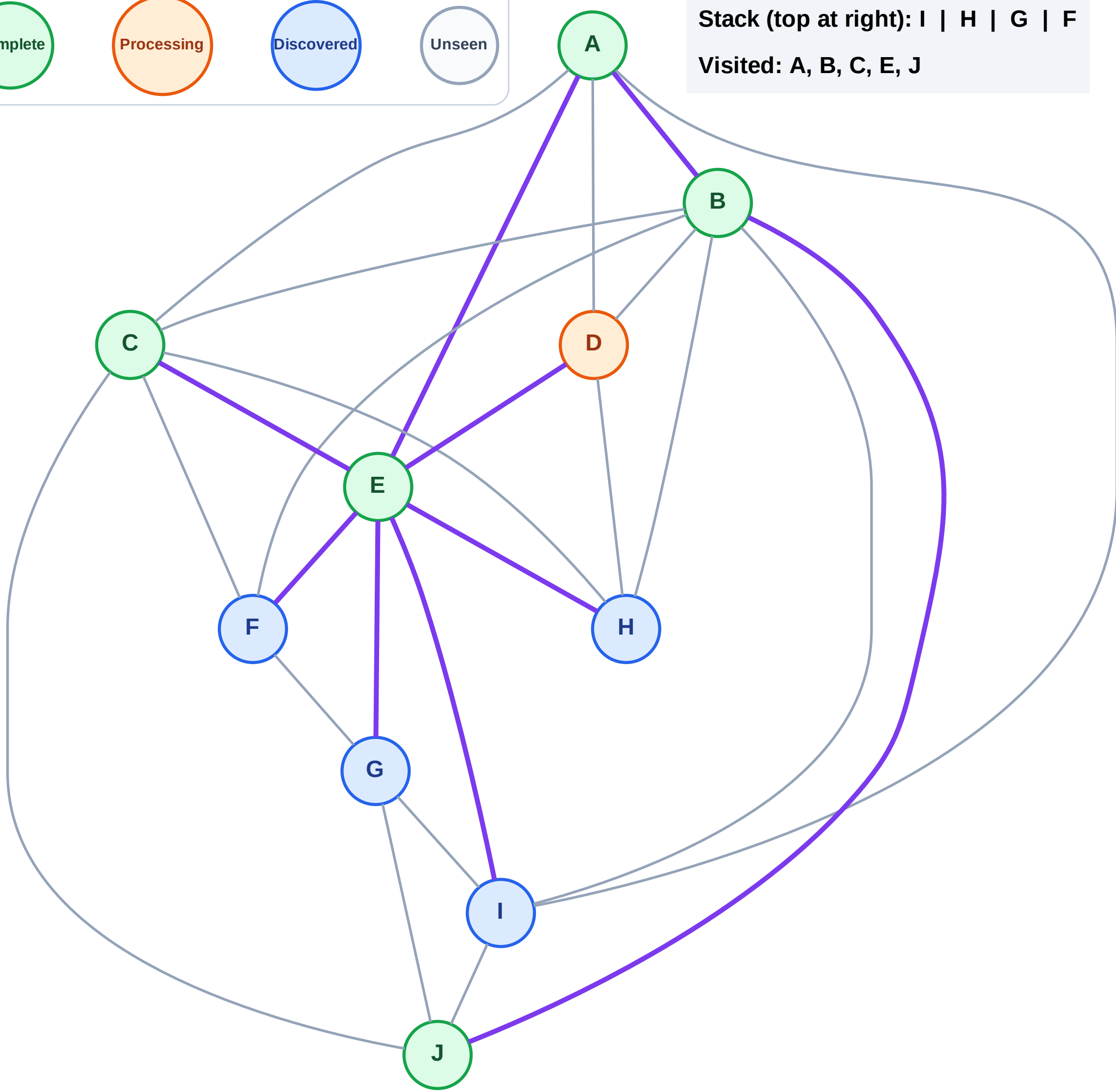
Step 12: Process node D

Legend



Stack (top at right): I | H | G | F

Visited: A, B, C, E, J



DFS Traversal

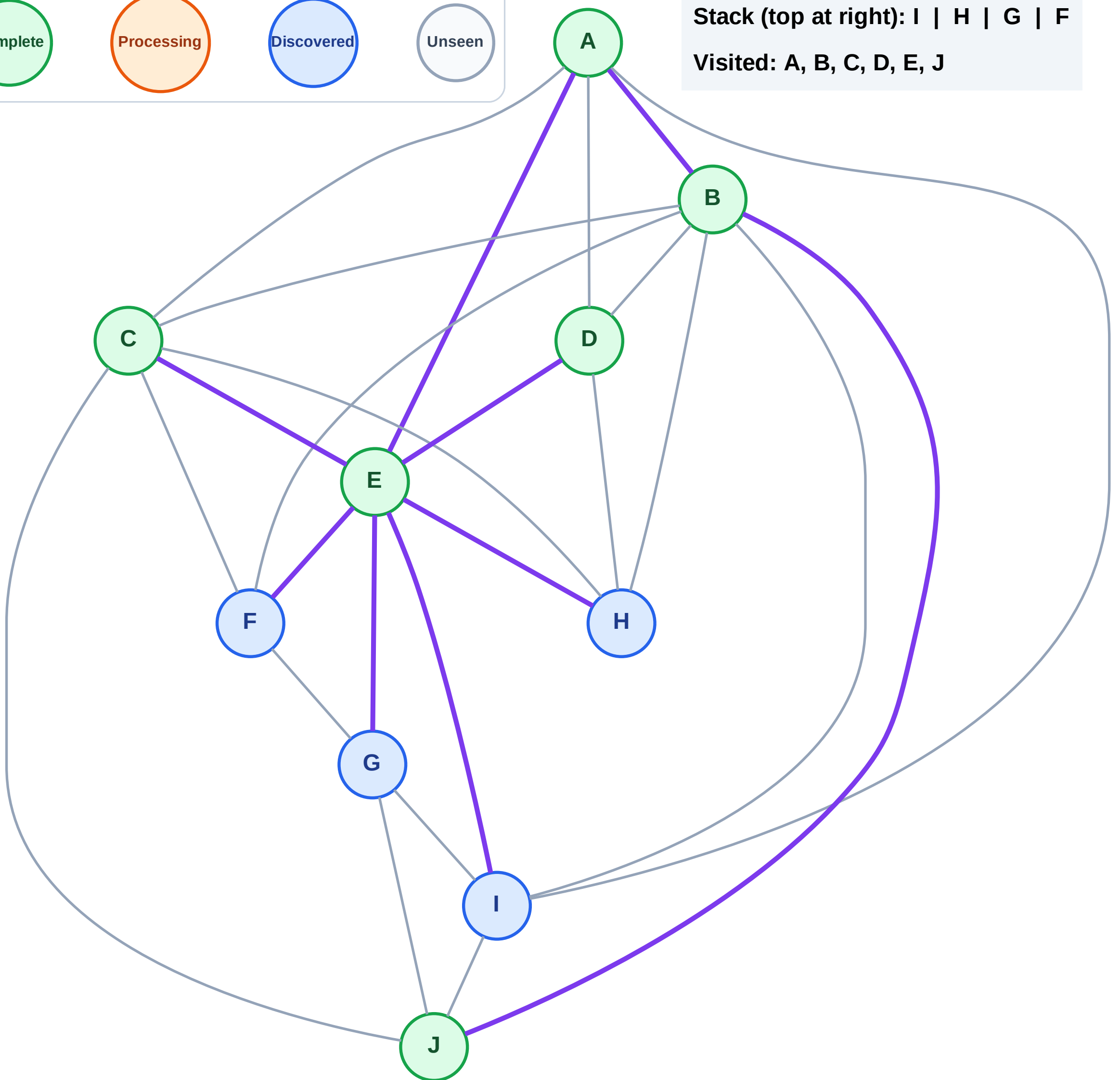
Step 13: No new neighbors from D

Legend

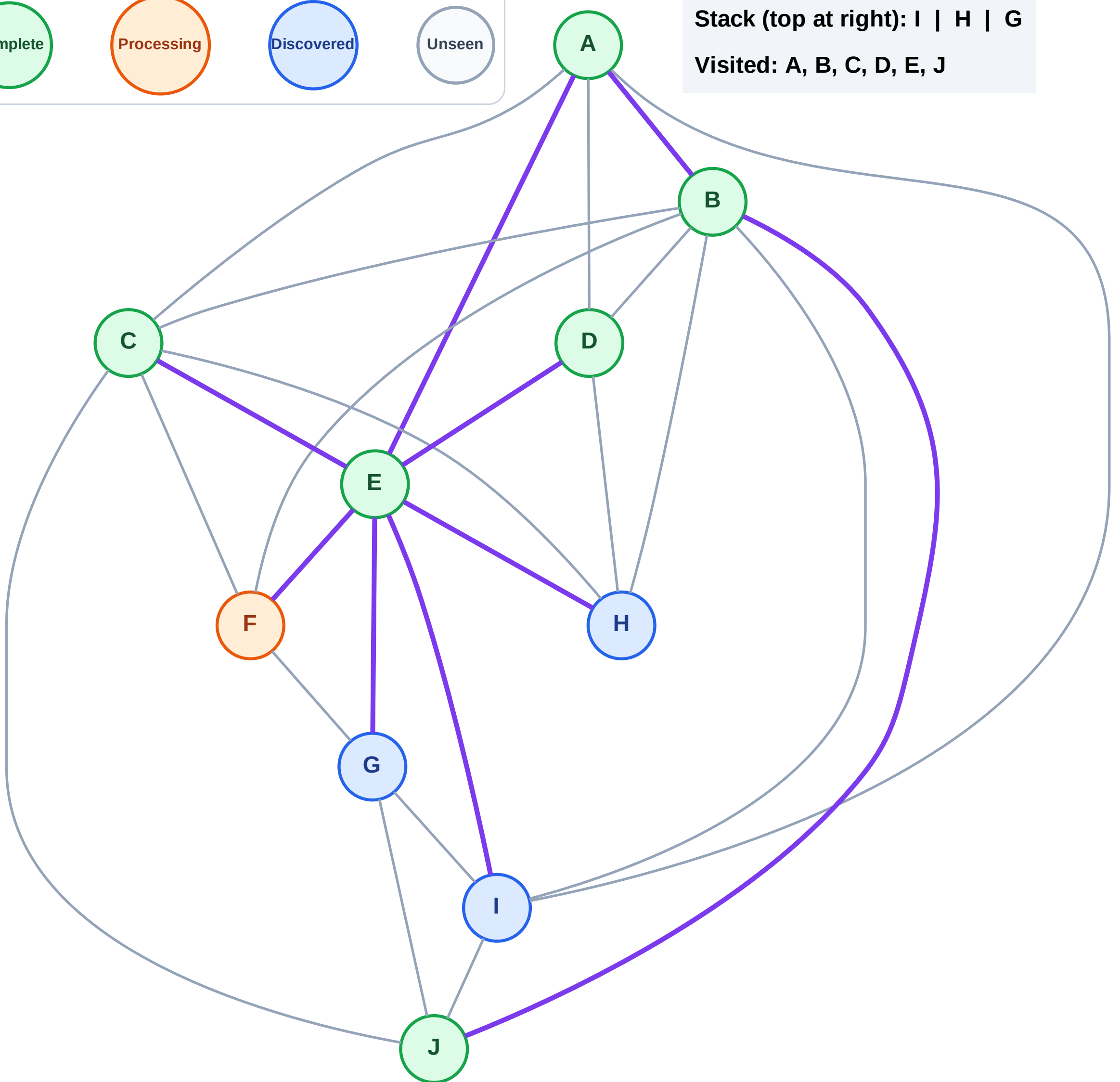
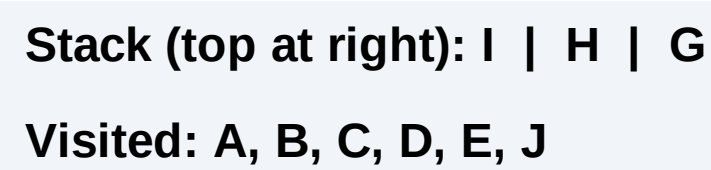


Stack (top at right): I | H | G | F

Visited: A, B, C, D, E, J



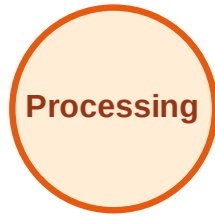
Step 14: Process node F



DFS Traversal

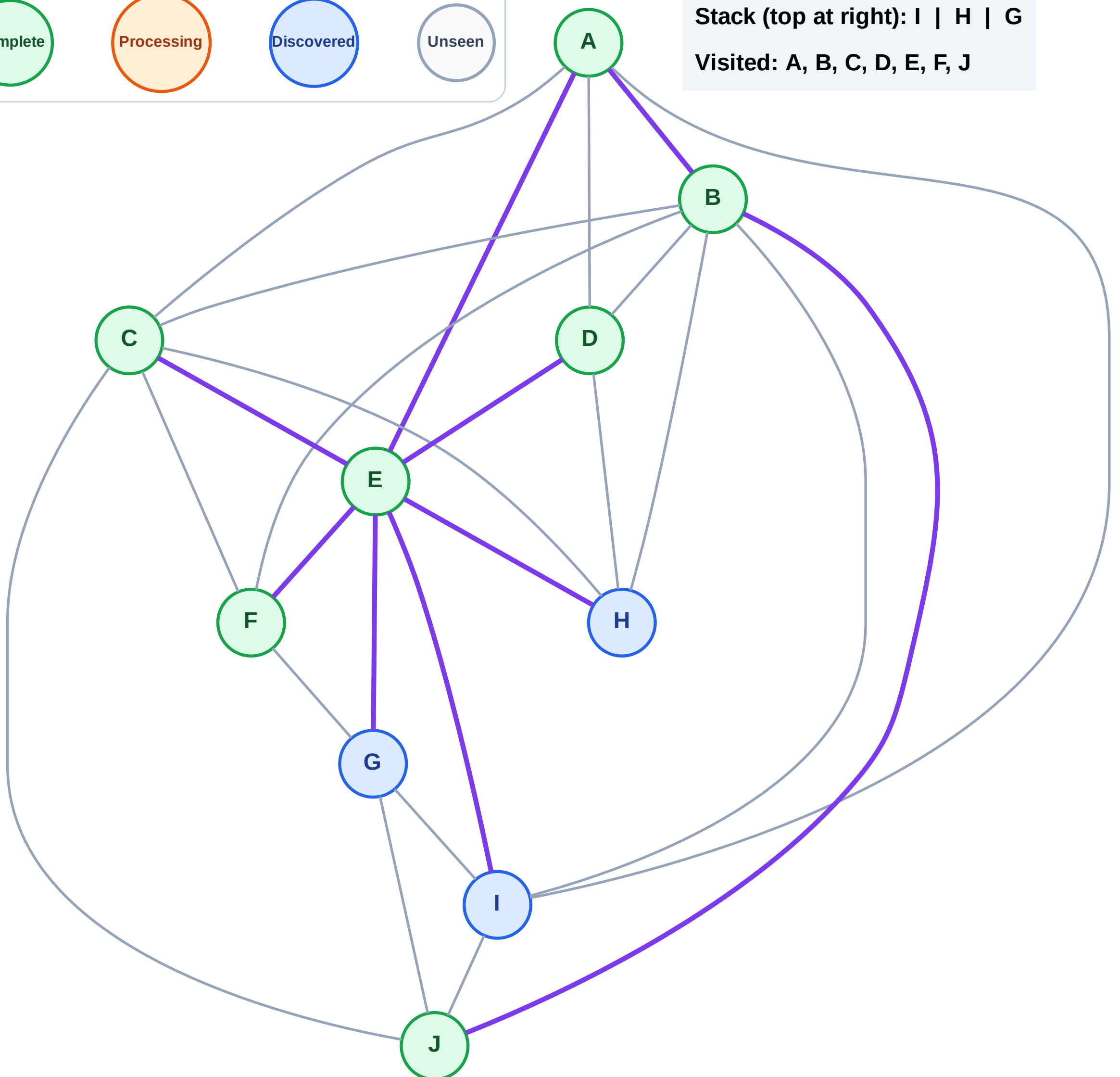
Step 15: No new neighbors from F

Legend



Stack (top at right): I | H | G

Visited: A, B, C, D, E, F, J



DFS Traversal

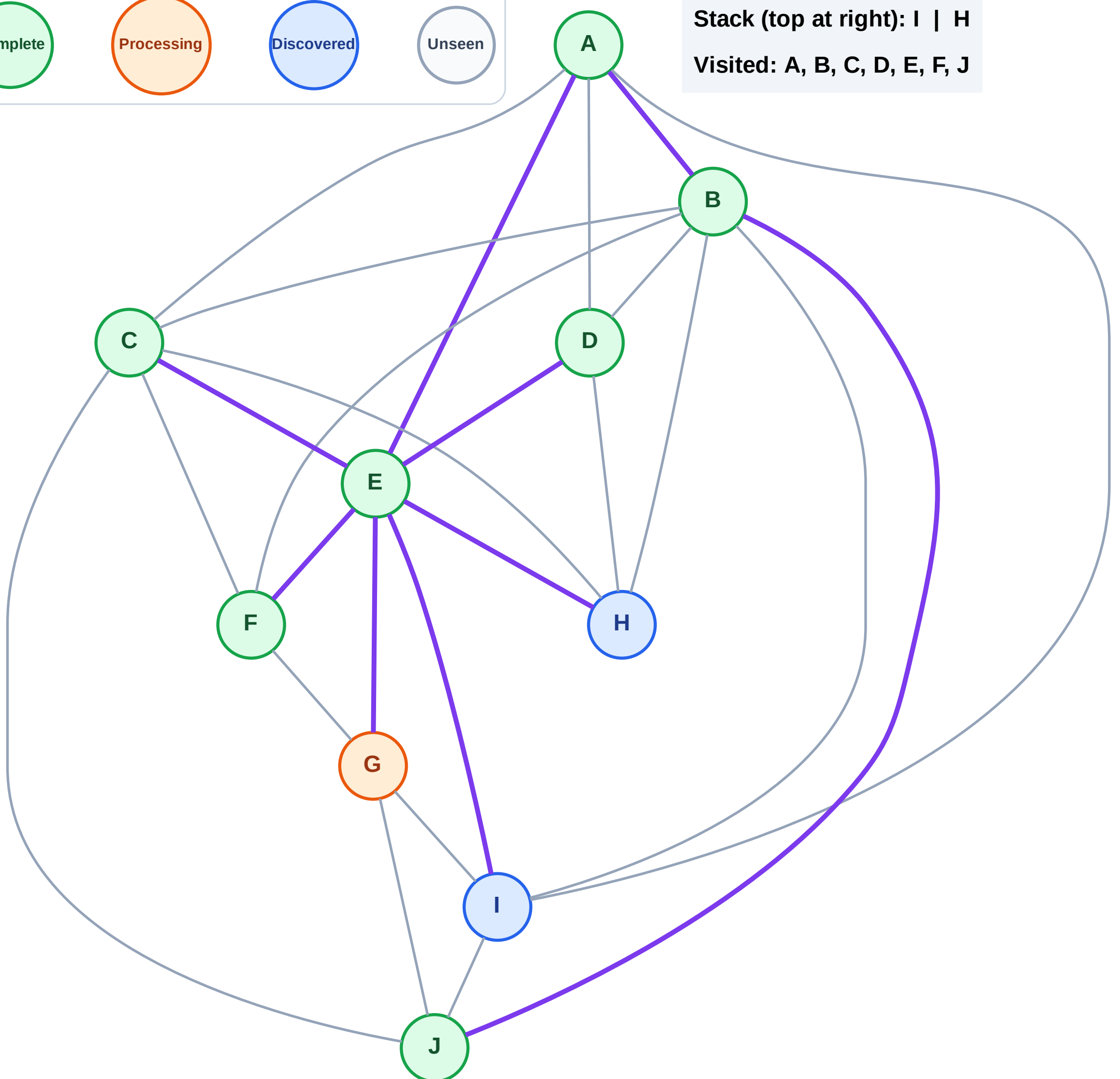
Step 16: Process node G

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, J



DFS Traversal

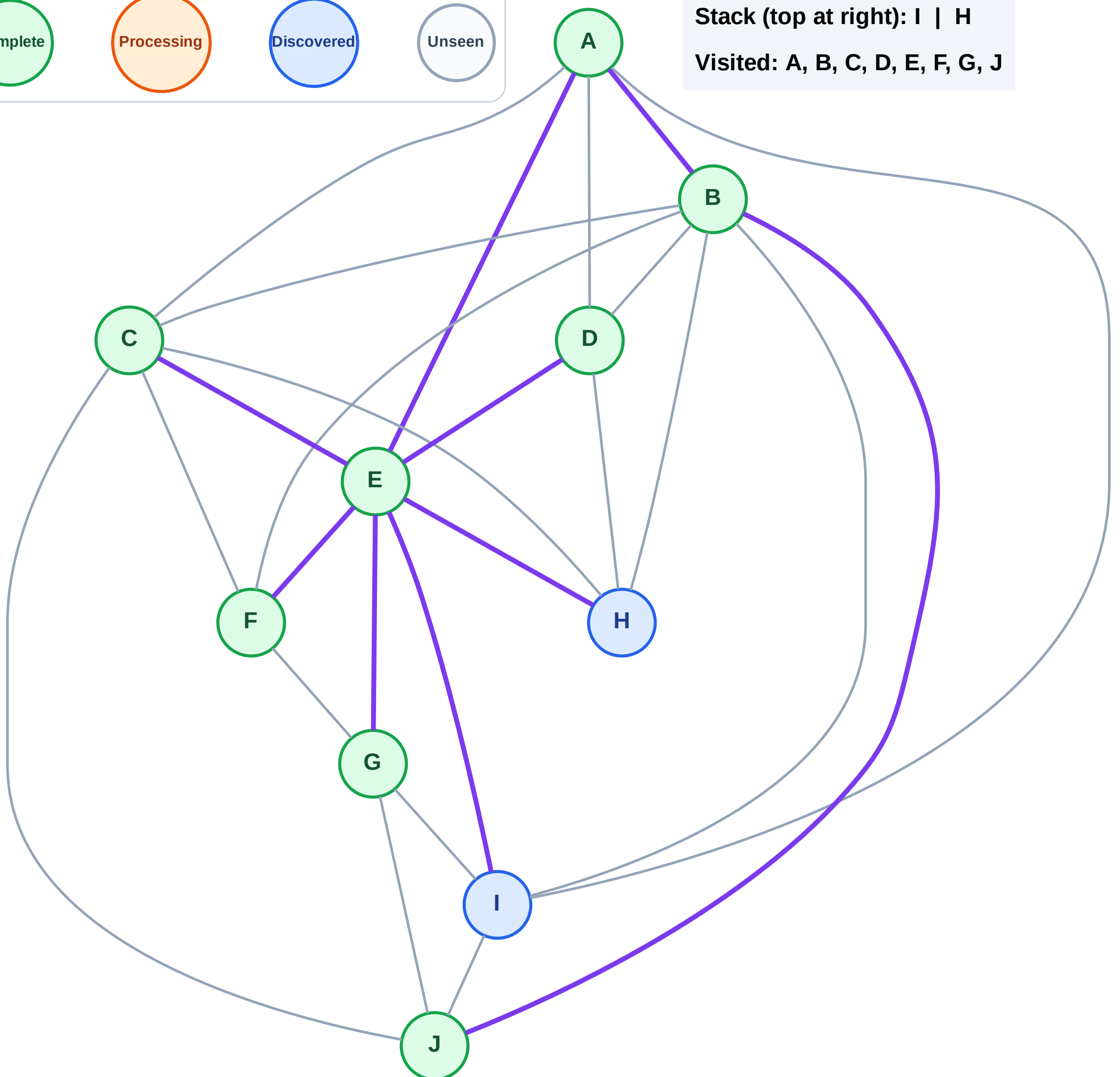
Step 17: No new neighbors from G

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

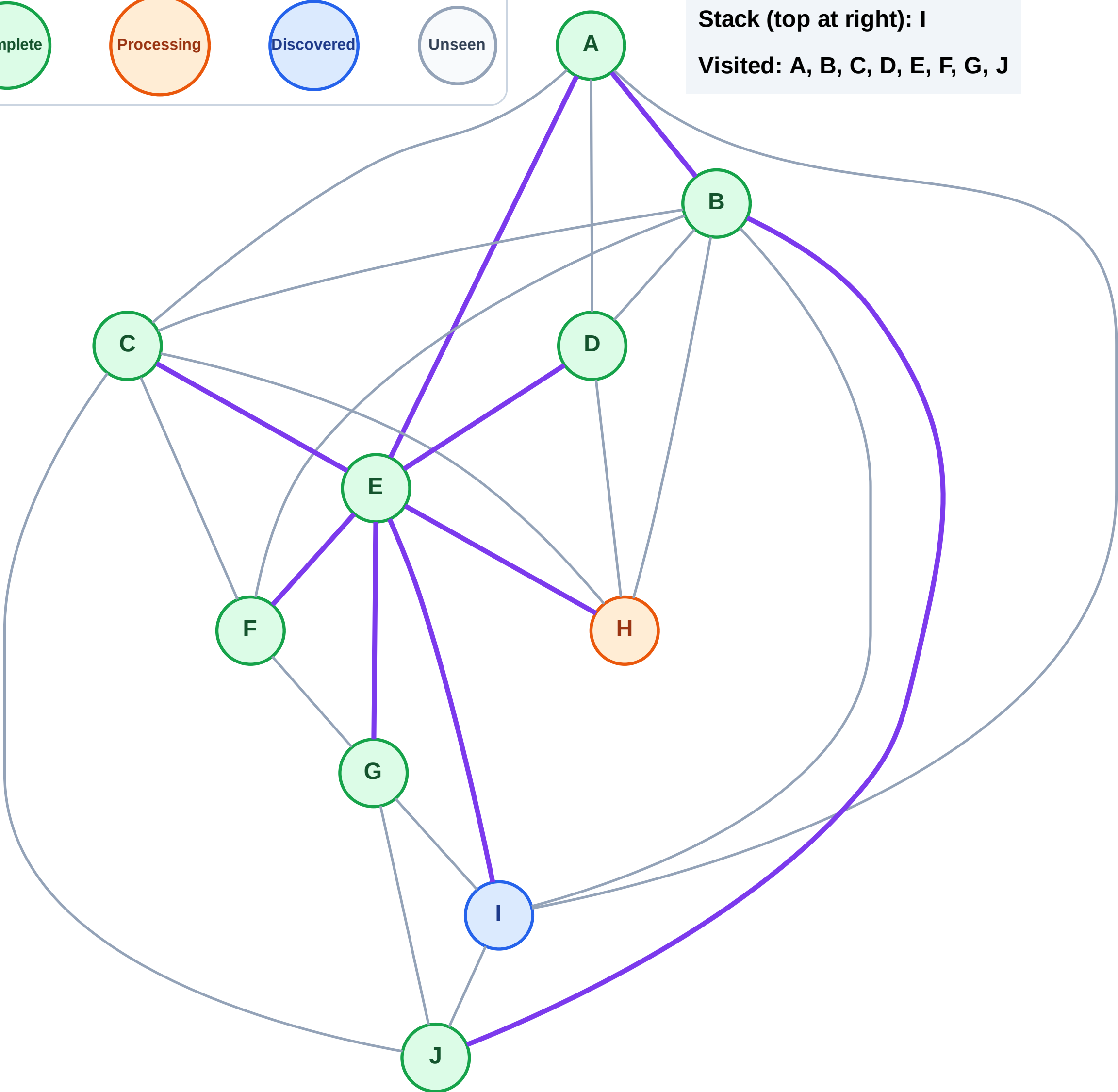
Step 18: Process node H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, J



DFS Traversal

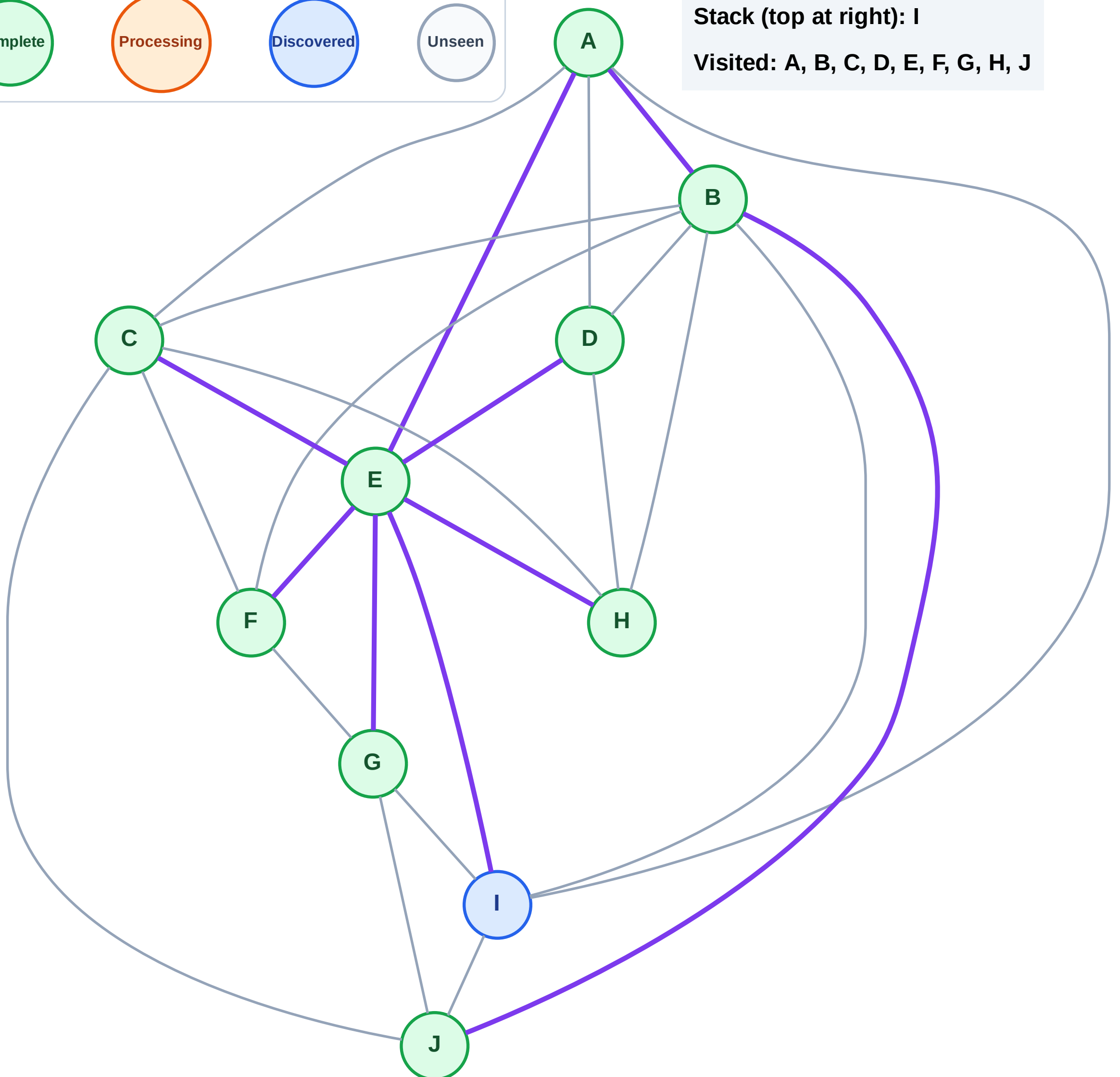
Step 19: No new neighbors from H

Legend



Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

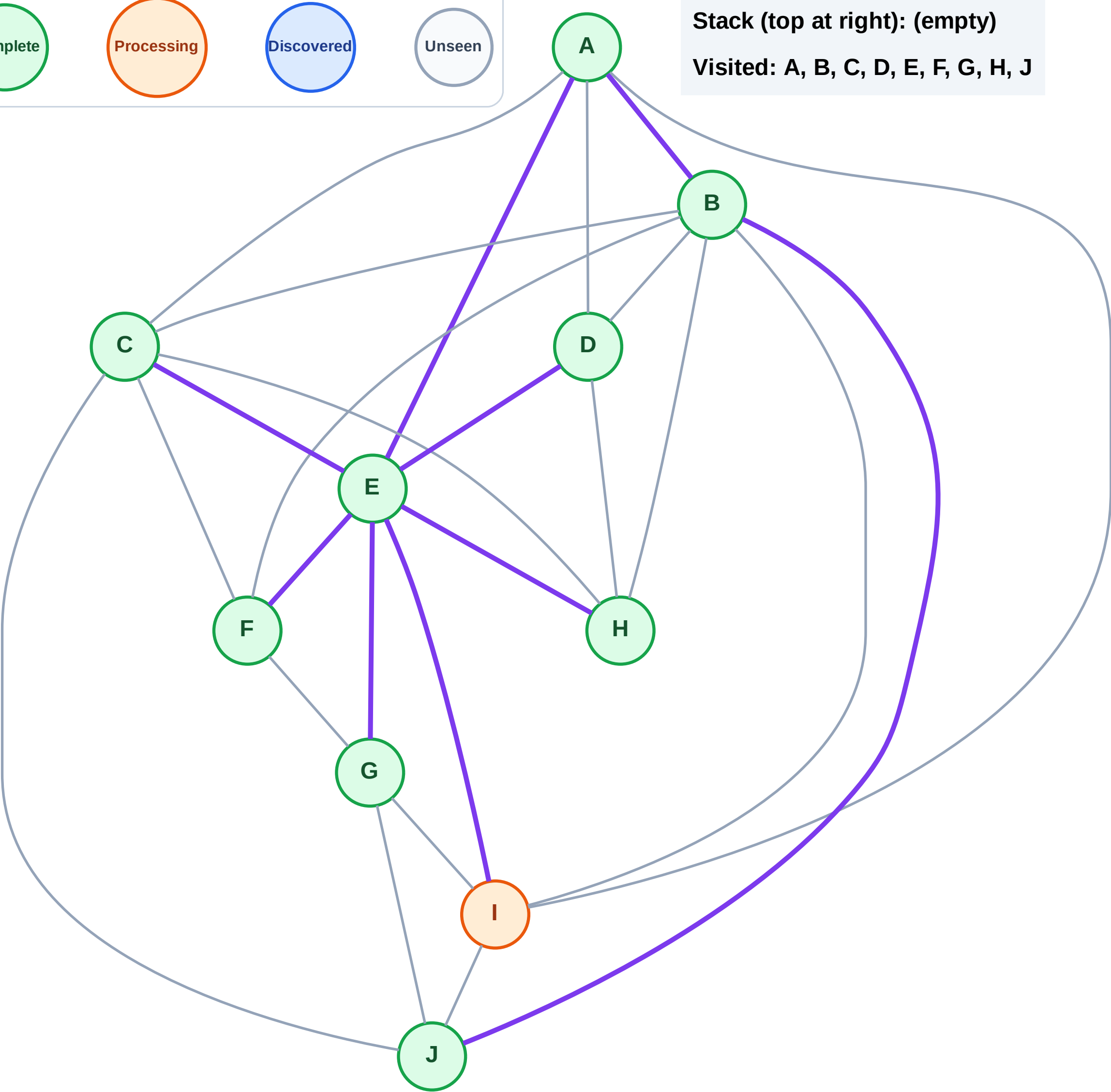
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

